

BizRadar v4

Universal Sector Algorithm Specification

Batch 5 — Final Batch (Standalone)

Sectors 51 (Information) + 61 (Education); Out-of-Scope sectors 55 + 92; Part 3 NAICS-4 Failure-Rate Database (302 codes); Part 4 Implementation Roadmap (5 phases with explicit gates); Part 5 Appendices (data sources, code structure, classifier keywords, glossary).

Read in conjunction with:

- Batch 1-2 PDF — Part 1 Architecture + Sectors 11/21/22/44-45/62/71/72/81 (114 pages)
- Batch 3 PDF — Sectors 23/52/53/54 (69 pages)
- Batch 4 PDF — Sectors 31-33/42/48-49/56 (62 pages)

This is the final batch of the v4 specification. With Batches 1-5 read together, engineering teams have the complete sector profile registry, failure-rate database, implementation roadmap, and appendices needed to ship v4 to production.

BATCH 5 CONTEXT — FINAL BATCH

Where Batch 5 fits

BizRadar v4 ships across 5 batches. Batches 1-4 covered 16 of the 18 in-scope NAICS sectors with named-entity depth. Batch 5 — this document — closes the spec by adding the remaining two specialty sectors (Information + Education), documenting the two out-of-scope sectors (55 + 92), and shipping the three operational artifacts engineering teams need: a 302-code failure-rate database, a 5-phase implementation roadmap with explicit gates, and four appendices covering data sources, code structure, classifier keywords, and glossary.

What is in this Batch 5 PDF

Section	Coverage
Ch. 51 — Information	4 sub-clusters: SaaS / media+publishing / telecom+ISP / broadcasting
Ch. 61 — Educational Services	4 sub-clusters: tutoring / K-12 private / bootcamps / trade schools
Out-of-Scope — Sectors 55 + 92	Holding companies + Public administration
Part 3 — Failure-Rate Database	302 NAICS-4 codes with y1/y3/y5/y10 survival rates
Part 4 — Implementation Roadmap	5 phases, explicit quality gates per phase
Part 5 — Appendices	Data sources, code structure, classifier keywords, glossary

Reading order recommendation

If you're an engineer about to ship v4 — read Part 4 Roadmap first to understand phase sequence and gate criteria, then Part 5 Appendix B for the sector profile registry structure, then Part 3 to understand failure rates by sector (informs prioritization). If you're a sector profile contributor — read your sector chapter first, then the closest-cousin sector chapter for substrate-reuse opportunities. If you're an executive — read the cover + Part 4 Roadmap + the high/low survivors lists in Part 3.

Quality bar — same as Batches 1-4

Every sub-cluster opportunity-playbook entry contains a NAMED real-world entity. SaaS chapter names Drata, Vanta, Outreach, Apollo, Gainsight, Vista Equity, Thoma Bravo, Salesforce AppExchange, Microsoft AppSource. Education chapter names ACCSC, ACCET, NAIS, NEASC, MSA, AMS Montessori, AWSNA Waldorf, CIRR, Title IV, Pell Grant, WIOA, GI Bill, named bootcamp ISA partners (Leif Capital, Vemo, Edly). Failure DB cites BLS BED + named industry sources (USDA NASS, EIA, FCC, NOAA NMFS) for every sector. Roadmap names every system component (Master Router, sectorProfile.js, businessTypeKeywordRouter.js, modelClassifier.js).

CHAPTER 51 — INFORMATION

SaaS + software, media + digital publishing, telecom + ISP, broadcasting + production. Phase 5 specialty — global radius, GTM-driven economics, valuation multiples nothing else in the spec touches.

One-line truth

A SaaS company has no physical radius — competes globally on G2 + Capterra + ProductHunt. A digital publisher has audience as inventory and ads + subscriptions + sponsorships as revenue, with zero overlap with retail logic. A local ISP runs network OpEx + capacity + named-carrier wholesale relationships. v3 has zero awareness of any of these.

NAICS code count: 35 **Sub-clusters:** 4

information.saas_software — SaaS + software companies (B2B SaaS, B2C apps, vertical SaaS, dev tools, AI products)

Business model	B2B_subscription + B2C_subscription
NAICS examples	511210, 518210
Modern variants	Vertical SaaS (HVAC, Legal, Healthcare), B2B PLG SaaS, AI-Native Product, Dev Tools / Open Source Commercial
Competitor radius	Global. SaaS competes worldwide on category. Vertical SaaS may have geographic moat by named-vertical regulator (e.g. legal SaaS state-by-state). API-first / dev tools: global with English + named-locale support.

Data sources

Source	Returns	Cost	Fallback
G2 + Capterra + GetApp + Software Advice	B2B software reviews + category leaders	Subscription / sub	Direct
Product Hunt + BetaList + Hacker News Show HN	Launch + early-traction signal	Free	Direct
Crunchbase + PitchBook + CB Insights	Funding + competitor + acquisition intel	Subscription	Public summary
LinkedIn + Apollo + ZoomInfo + Clay	B2B prospect + buyer intel	Subscription	LinkedIn
SimilarWeb + Semrush + Ahrefs	Competitor traffic + SEO + paid intel	Subscription	Direct
Built With + Wappalyzer + StackShare	Tech stack intel on prospects	Subscription / free	Direct

Source	Returns	Cost	Fallback
Gartner + Forrester + IDC + 451 Research	Analyst-tier coverage + waves + magic quadrants	Subscription	Public summary
SaaStr + ProfitWell + Maxio + ChartMogul benchmarks	SaaS metric benchmarks (NRR, GRR, CAC, LTV, magic number)	Subscription / public	Direct

Review categories tracked

- Product fit + workflow alignment
- Reliability + uptime
- Onboarding + time-to-value
- Customer support / response
- Pricing transparency vs ROI
- Integration ecosystem (named platforms)
- Roadmap velocity + new-feature flow
- Security + compliance posture (SOC 2, HIPAA, GDPR)

Opportunity playbook (15 categories)

Category	Research basis	Example with named entity
Product-Led Growth (PLG) motion + free tier / freemium	Named PLG playbooks (Notion, Slack, Figma, Atlassian, Calendly economics)	Free tier + PLG motion + named usage-trigger (e.g. seat or usage cap) → drives 4-8% free-to-paid conversion at \$0 sales cost vs sales-led \$148 CAC
G2 + Capterra category leadership pursuit	Named category + review velocity + G2 grid criteria	G2 Leader / Momentum Leader badge in named category + 200+ reviews — drives 28% of inbound demos at \$0 ad spend
SOC 2 Type 2 + ISO 27001 + named compliance	Named auditors (Drata, Vanta, Secureframe, Tugboat Logic, A-LIGN, Schellman, Coalfire)	SOC 2 Type 2 via Drata + named audit firm — unlocks enterprise + named-account RFPs requiring proof of compliance
Vertical SaaS pivot (industry specialization)	Named vertical SaaS exits + valuation premiums	Vertical SaaS for named industry (HVAC, dental, legal, construction) — 8-12× ARR vs horizontal 4-6× ARR + lower competition + higher NRR
AI / LLM product augmentation	Named AI integration patterns (OpenAI API, Anthropic Claude API, Pinecone, Llama)	AI-native feature integrated via OpenAI / Anthropic API — captures premium pricing + competitive moat + reactivates churned users
Outbound + ABM + intent-data motion	Named tools (Outreach, SalesLoft, Apollo, Clay, 6sense, Demandbase, Bombora intent)	Apollo + Outreach + 6sense ABM on 200 named target accounts — drives 8-14 qualified meetings/mo at \$148 CAC

Category	Research basis	Example with named entity
Channel + reseller + named partnership	Named programs (HubSpot, Salesforce AppExchange, Microsoft AppSource, AWS Marketplace)	AppExchange or AWS Marketplace listing + named SI partner — drives 18-28% of new ARR through partner channel
Content + SEO + thought leadership engine	Named platforms (Ahrefs, Semrush) + content economics	Topic-cluster content strategy + 18 mo runway — drives 14k organic monthly visitors + 8% of pipeline at \$0 ongoing cost
Annual contract + multi-year + auto-renewal	Annual vs monthly + cohort retention impact	Annual contract motion + named CRM (HubSpot or Salesforce) discipline — drops gross churn from 18% to 7% + improves NRR
Pricing + packaging optimization	Named pricing frameworks (Patrick Campbell, ProfitWell, Simon-Kucher)	Pricing study + value metric refinement — captures 14-22% revenue lift on existing book + improves expansion
Customer success + expansion motion	Named CS platforms (Gainsight, Catalyst, Vitally, Totango, ChurnZero)	Named CS platform + dedicated CSM team — drives NRR from 105% to 128% + drops gross churn
Strategic acquisition + roll-up	Named M&A; advisors + valuation multiples (4-15× ARR depending on metrics)	Acquire complementary product at 6× ARR — drives revenue + cross-sell + product expansion + defensive moat
API + developer + open-source motion	Named API + dev-tool strategies	API-first + open-source strategy + named developer community — drives bottoms-up enterprise adoption + premium pricing
Capital strategy (bootstrap vs venture vs PE)	Named investors + valuation benchmarks	Series A or B raise at named-fund — accelerates GTM + product roadmap, OR PE recap for liquidity + capital discipline
International + named-locale expansion	Named localization platforms + payment + tax (Stripe Tax, Avalara, Lokalise, Crowdin)	EU + UK + AU expansion via Stripe Tax + named localization — captures 22-38% revenue lift over 18 mo

Specificity rules (anti-hallucination)

- Every named platform must be real (Stripe, Stripe Tax, Avalara, Drata, Vanta, Secureframe, A-LIGN, Schellman, Coalfire, Outreach, SalesLoft, Apollo.io, Clay, 6sense, Demandbase, Gainsight, Catalyst, Vitally, Totango, ChurnZero)
- Every named SaaS metric must follow real benchmarks (NRR, GRR, CAC, CAC payback, LTV, magic number, Rule of 40, ARR growth, gross margin)

- Every named compliance framework must be real (SOC 2 Type 1 + Type 2, ISO 27001, HIPAA, GDPR, CCPA, PCI-DSS, FedRAMP, StateRAMP)
- Every named acquirer / investor must be real (Vista Equity, Thoma Bravo, Insight Partners, Bessemer, Sequoia, a16z, Tiger Global, Tidemark, Iconiq)
- Every named partner / marketplace must be real (Salesforce AppExchange, HubSpot Marketplace, Microsoft AppSource, AWS Marketplace, GCP Marketplace, Atlassian Marketplace, Shopify App Store)

Cost scale calibration

SaaS + software spend bands operate at people + GTM + cloud-infrastructure scale:

Band	Spend type
1 — Under \$10,000	Tool license, founder-led campaign, content batch
2 — \$10,000 to \$100,000	GTM tool stack, content engine, contractor team
3 — \$100,000 to \$1M	Sales hire, paid acquisition, compliance audit, vertical specialty
4 — \$1M to \$10M	Multi-team scale, multi-region expansion, M&A; small
5 — Over \$10M	Major M&A;, multi-product expansion, enterprise GTM build

Persona schema

B2B (decision committee) + B2C (individual buyer) hybrid

- Buyer role (champion + economic buyer + IT/security + procurement for B2B; individual for B2C)
- Buying motion (PLG bottoms-up / sales-led top-down / hybrid)
- Decision driver (functionality, integration, price, security, support, vendor stability)
- Vertical / industry match
- Acquisition channel (organic, paid, content, marketplace, referral, partner, outbound)
- LTV ($ARR \times NRR \times \text{retention years} \times \text{cross-sell expansion}$)

Cycle / calendar model

SaaS runs on calendar + budget + renewal + product-cycle:

Window	Trigger	Action
Q4 (Oct-Dec)	Budget approval + year-end push	Pipeline build for Q1 + renewal discipline
Q1 (Jan-Mar)	New-year deployments + budget execution	New-account ramp + onboarding
Annual: customer renewal cycle	Multi-year + annual auto-renewal	Renewal motion + expansion CSM discipline
Quarterly: customer QBR + expansion review	Health-score driven	QBR cadence + expansion-opportunity surfacing
Product release cadence (2-week sprints)	Continuous product flow	Release notes + customer comm + feature adoption

Window	Trigger	Action
Annual: SOC 2 / cert audit cycle	Compliance + audit	Pre-audit prep + control evidence
Capital cycle (18-24 mo)	Funding rounds or PE re-cap	Metrics + investor update + raise prep

Issue library

- Customer churn / NRR slipping — universal — frame as CSM + named platform + expansion motion
- CAC payback lengthening — universal — frame as PLG + content + ABM + segment focus
- Compliance burden growing (SOC 2, HIPAA, FedRAMP) — universal — frame as Drata / Vanta automation
- Product-roadmap velocity vs scaling team — universal — frame as eng productivity + AI augmentation
- Channel + partner mix vs direct — universal — frame as named partner program + co-sell motion
- Pricing pressure + commoditization — universal — frame as vertical specialty + ROI articulation
- Multi-product transition + cross-sell — universal — frame as bundle + expansion motion
- Capital strategy (bootstrap vs raise vs sell) — universal — frame as Rule of 40 + named-investor + acquirer

Report sections

Section	Contents
Company snapshot	ARR, customer count, headcount, segment + vertical mix, named comp set
Top 5 strategic moves	Drawn from 15-category playbook
GTM motion + channel optimization	PLG / sales-led / hybrid + named tools
Compliance + security posture roadmap	Named programs + audit prep
Product + AI + roadmap velocity plan	Named integrations + capability priorities
Customer success + NRR plan	Named CS platform + expansion motion
Capital + M&A; strategy	Named investors / acquirers + valuation roadmap
Known industry challenges	Collapsed drawer

KPIs to track

- ARR + ARR growth %
- Net Revenue Retention (NRR target > 110% B2B)
- Gross Revenue Retention (GRR target > 90%)
- CAC + CAC payback (target < 18 mo)
- Magic number + Rule of 40
- Gross margin % (target > 70% B2B SaaS)

- Logo + dollar churn
- Pipeline coverage (target 3-4× quota)

information.media_publishing — Media + digital publishing (newspapers, magazines, blogs, podcasts, newsletters, content sites)

Business model	B2C_subscription + B2B_advertising hybrid
NAICS examples	511110, 511120, 511130, 519130
Modern variants	Substack Newsletter, Premium Newsletter Network, Podcast Network, Niche Trade Publication
Competitor radius	Niche / topic-driven: global. Local news: 30-100 mi metro. Trade publication: industry-defined. Newsletter: language + niche-defined. Podcast: global within language.

Data sources

Source	Returns	Cost	Fallback
Google Analytics + Google Search Console	Audience + traffic + SEO performance	Free	Direct
SimilarWeb + Semrush + Ahrefs	Competitor + search intel	Subscription	Direct
Substack + Beehiiv + ConvertKit + Ghost	Newsletter platform benchmarks	Subscription / public	Direct
Podtrac + Spotify for Podcasters + Apple Podcasts Connect	Podcast download + listener intel	Free / sub	Direct
Press Gazette + Digiday + Nieman Lab	Industry trends + benchmarks	Subscription	Public
Named ad networks + sponsorship platforms (Google AdSense, Carbon Ads, BuySellAds, Mediavine, Raptive)	Ad rev + sponsorship benchmarks	Direct	Industry
Memberful + Patreon + Stripe Billing	Subscription + membership economics	Subscription	Direct
Local newspaper association + INMA + RJI	Industry benchmarks + named peers	Member-only	Public

Review categories tracked

- Editorial quality + voice

- Content frequency + consistency
- Subscription value + paywall fairness
- Site / app experience + load speed
- Sponsorship transparency
- Comment + community quality
- Mobile + email experience
- Niche / category authority

Opportunity playbook (14 categories)

Category	Research basis	Example with named entity
Newsletter as flagship product (vs site)	Named newsletter playbooks (Morning Brew, The Hustle, Stratechery, Lenny's Newsletter)	Newsletter pivot via Substack / Beehiiv / Ghost — drives 60-80% of revenue from email + sponsor mix at higher CTR + LTV vs banner display
Subscription / paywall + premium tier	Named platforms (Memberful, Substack Pro, Pico, Tinypass / Piano, Outline)	Hard paywall + premium tier \$14/mo × 4,800 subs = \$806k/yr recurring at 80% margin (vs ad-only \$0.04 RPM)
Niche specialization + authority	Named niche publications + audience-revenue benchmarks	Niche category authority (e.g. craft brewing, parenting, finance) — captures premium \$/sponsorship + sponsor-direct vs network
Sponsorship + direct ad sales (vs network)	Named platforms (BuySellAds, Carbon Ads, Sponsored Post Sales)	Direct sponsorship \$4-12k per campaign + 4 partners — drives \$250-600k/yr at 90% margin (vs ad-network 50% of \$0.04 RPM)
Programmatic + premium ad ops + named ad ops platform	Named platforms (Mediavine, Raptive, AdThrive, Sovrn, Magnite)	Mediavine or Raptive partnership — captures 4× display RPM vs Google AdSense (typical \$0.04 → \$0.18 RPM)
Podcast launch + monetization	Named platforms (Acast, Megaphone, Buzzsprout, Anchor, Captivate, Pod Movement)	Branded podcast + named host-read sponsorship — adds \$14-48k/mo revenue + drives flagship audience growth
Events + conferences + summit revenue	Named event-tech platforms (Hopin, Bizzabo, Eventbrite, Hubilo)	Annual paid summit + named sponsors — adds \$400-1.4M revenue at 38% margin + drives subscription + community
Membership + community line	Named community platforms (Discourse, Circle, Mighty Networks, Slack, Discord)	Premium community membership \$14-48/mo — drives recurring revenue + retention moat + sponsor visibility

Category	Research basis	Example with named entity
SEO + topic-cluster content engine	Named SEO platforms + content velocity	Topic-cluster strategy + 4 articles/wk over 18 mo — drives 80k organic monthly visitors + drives ad + subscriber funnel
AI / automation in content production	Named platforms (Jasper, Claude, ChatGPT, Anthropic API, Otter.ai for transcription)	AI-augmented content + transcription + named editorial review — drops cost per published piece 60% + scales coverage
Acquisition / consolidation	Named M&A; advisors + valuation multiples (3-6× revenue, 4-8× EBITDA)	Acquire complementary niche publication or newsletter — adds audience + cross-sell + ad inventory
Affiliate + commerce + reader-revenue line	Named programs (Amazon Associates, Skimlinks, Awin, Impact, ShareASale)	Skimlinks + named affiliate networks + curated commerce — adds \$14-48k/mo at 100% margin from existing audience
Audio + video + multi-format content distribution	Named distribution + platforms	YouTube + TikTok + LinkedIn + podcast multi-format strategy — drives discovery + audience growth across surfaces
Reader-supported + crowdfunding + named-reader revenue	Named platforms (Patreon, Buy Me a Coffee, Memberful)	Reader-supported model + Patreon + named tier — drives recurring revenue from highest-engaged readers

Specificity rules (anti-hallucination)

- Every named platform must be real (Substack, Beehiiv, ConvertKit, Ghost, Memberful, Patreon, Pico, Tinypass / Piano, Outline)
- Every named ad platform must be real (Google AdSense, Carbon Ads, BuySellAds, Mediavine, Raptive, AdThrive, Sovrn, Magnite, Pubmatic)
- Every named podcast platform must be real (Acast, Megaphone, Buzzsprout, Anchor, Captivate, Spotify for Podcasters)
- Every named association must be real (INMA International News Media Association, RJI Reynolds Journalism Institute, AAM Alliance for Audited Media)
- Every named publication referenced must be real and currently operating

Cost scale calibration

Media + publishing spend bands operate at content + audience + ad-stack scale:

Band	Spend type
1 — Under \$1,000	Platform setup, branded template, single newsletter launch
2 — \$1,000 to \$10,000	Editorial freelance, ad-stack setup, podcast equipment
3 — \$10,000 to \$100,000	Editorial team build, paywall, sponsor + sales team
4 — \$100,000 to \$1M	Multi-format expansion, M&A; small, events launch

Band	Spend type
5 — Over \$1M	Major M&A;, multi-vertical expansion, enterprise content ops

Persona schema

B2C (reader/subscriber) + B2B (sponsor/advertiser) hybrid

- Reader type (free / casual / premium subscriber / community member)
- Acquisition channel (SEO, social, newsletter cross-promo, podcast, referral)
- Engagement tier (single visit / regular / paying / community)
- Sponsor type (brand sponsor / endemic sponsor / programmatic / direct)
- Decision driver for paying (quality, niche authority, community, exclusivity)
- LTV (subscription years × ARPU + community + commerce)

Cycle / calendar model

Media + publishing run on news cycle + season + sponsor budget cycles:

Window	Trigger	Action
Q4 holiday + year-end (Oct-Dec)	Sponsor budget peak + holiday content	Maximum sponsor revenue + content velocity
Q1 (Jan-Mar)	New-year audience + subscription push	Subscription drive + new-content launch
Summer (Jun-Aug)	Slower B2B + high B2C	Lifestyle content peak + strategic content build
Annual: subscription renewal cycles	Subscriber retention discipline	Renewal comm + retention motion
Daily / weekly: editorial cadence	Content production	Content discipline + quality consistency
Annual: ad rate card + renewal	Sponsor + advertiser cycles	Rate-card discipline + sponsor retention
Quarterly: audience + analytics review	Audience health check	Analytics + content-mix optimization

Issue library

- Subscriber churn vs acquisition cost — universal — frame as content quality + retention motion
- Programmatic ad RPM erosion — universal — frame as named premium ad ops + direct sales
- Sponsor + advertiser dependency on few named accounts — universal — frame as diversification + sponsor pipeline
- Editorial team velocity vs cost — universal — frame as AI augmentation + freelance + named editorial discipline
- SEO algorithm shifts (Google updates) — universal — frame as authority + brand + diversified channel mix

- Platform dependency (Twitter, TikTok, etc.) — universal — frame as owned audience + email + community
- Audience growth plateau — universal — frame as new format + named partnership + niche specialization
- Monetization stack complexity — universal — frame as named platforms + simplified rev mix

Report sections

Section	Contents
Publication snapshot	Audience, content cadence, revenue mix, named comp set
Top 5 monetization + audience moves	Drawn from 14-category playbook
Subscription + membership tier plan	Named platforms + tier architecture
Sponsorship + direct ad pipeline	Named sponsor prospects + direct sales structure
Multi-format + multi-platform strategy	Newsletter / podcast / video / events roadmap
SEO + audience growth engine	Named tools + topic-cluster strategy
Reader personas (4 + lost subscriber)	Free / casual / paying / community / churned
Known industry challenges	Collapsed drawer

KPIs to track

- Audience size + monthly unique visitors
- Email list size + open rate + CTR
- Subscription count + retention + ARPU
- Ad / sponsorship revenue + RPM
- Content velocity (pieces / week)
- SEO authority + organic traffic %
- Diversification (% revenue from largest 3 sources)
- Gross margin % (target > 60% digital pub)

information.telecom_isp — Telecom + ISP (regional / rural ISP, fixed wireless, fiber, VoIP carrier, MVNO)

Business model	B2C_subscription + B2B_subscription
NAICS examples	517311, 517312, 517410, 517919

Competitor radius	Local + regional service area defined by network footprint. Wireline / fiber: defined by named footprint. Fixed wireless: 5-25 mi tower coverage. Cable / DSL incumbents (Comcast, Spectrum, Cox, AT&T, Verizon, CenturyLink, Frontier) compete in every metro.
--------------------------	---

Data sources

Source	Returns	Cost	Fallback
FCC Broadband Map + Form 477 + BDC	Broadband coverage + named competitor footprint	Free public	Direct
BroadbandNow + HighSpeedInternet.com	Consumer-side competitive intel	Free / sub	Direct
NTCA + WISPA + FBA + ITTA	Industry benchmarks + named peers	Member-only	Public
Named OSS / BSS platforms (Sonar, ITRoam, Sandvine, Calix, RouteCo)	Internal benchmarks + tools	Subscription	Direct
NTIA + state broadband office + USDA RUS	Federal + state broadband funding programs	Free public	Direct
Named wholesale carriers (Crown Castle, Zayo, Lumen, Windstream, Frontier, GTT)	Wholesale connectivity intel	Direct	Industry
Local ILEC + cable competitor data	Named regional competitor intel	Free / FCC	Direct
Named CRM + billing (Sonar, Powercode, AzotelOSS, Sandvine)	Internal benchmarks	Subscription	Direct

Review categories tracked

- Speed + reliability (vs advertised)
- Latency + packet loss
- Customer service / responsiveness
- Outage handling + comms
- Pricing transparency + no-fee culture
- Installation experience
- Coverage + serviceability
- Support of named services (gaming, streaming, work-from-home)

Opportunity playbook (13 categories)

Category	Research basis	Example with named entity
BEAD (Broadband Equity Access Deployment) program	NTIA BEAD + state-level allocation programs	BEAD application via state broadband office — captures \$10-100M/yr deployment grant in unserved + underserved areas
USDA ReConnect + RUS loan + grant programs	Named USDA programs + ReConnect rounds	USDA ReConnect Round 5 application — \$4-50M grant + loan combo for rural fiber buildout
State broadband office + named funding programs	State-level broadband programs	State broadband fund + named program (varies by state — e.g. Minnesota Border-to-Border, NY Broadband Program, Wisconsin) — captures matching grant
Fiber-to-the-home (FTTH) buildout	Named FTTH economics + named equipment partners (Calix, Adtran, Nokia, Tibit Communications)	Named area FTTH buildout + 2,000 passings + 50% take rate + \$80/mo ARPU = \$80k/mo recurring revenue at 65% gross margin
Fixed wireless (FW) expansion	Named FW equipment partners (Cambium, Mimosa, Tarana, Ubiquiti)	Tarana G1 + 10-mile coverage rings — drives premium \$/sub vs DSL + low capex per home passed (vs FTTH)
Business + enterprise data services	Named B2B services (dedicated fiber, SD-WAN, cloud connectivity)	Dedicated business fiber + SD-WAN partnerships (named providers Versa, Cato, Cisco Meraki, Fortinet) — adds \$1.4-4.8k/mo per business customer at 70% margin
OSS / BSS modernization	Named platforms (Sonar, ITRoam, Powercode, Sandvine, Calix Cloud, RouteCo, Sandvine, Splynx)	Sonar OSS + automated provisioning + customer self-service — drops admin time 40% + drives customer portal adoption + improves NPS
Customer experience + Net Promoter Score	Named NPS programs + customer comm	NPS measurement + service quality program — drives customer retention + premium pricing in named markets
IPTV / streaming bundle	Named IPTV platforms (BendTV, Innovative Systems, MobiTV, named local streaming)	IPTV bundle for fiber subs — drives ARPU lift \$14-48/mo + drops churn
VoIP + business voice + UCaaS	Named VoIP / UCaaS partners (8x8, RingCentral, Vonage Business, Microsoft Teams Phone, Zoom Phone)	White-label or partner UCaaS — adds \$14-32/seat/mo recurring at 50% margin

Category	Research basis	Example with named entity
Wholesale + carrier-of-carriers business	Named wholesale opportunities + carrier hotels	Wholesale capacity + transit + dark fiber + named carrier hotels — drives wholesale ARPU at 28% margin
Named-tower partnership + colocation	Named tower partners (American Tower, Crown Castle, SBA Communications)	Tower lease + named tower partner colocation — drops capex on FW + small-cell deployment + accelerates coverage
M&A; + ISP roll-up strategy	Named M&A; advisors + valuation multiples (8-14× EBITDA + per-pass \$300-2,400)	Acquire smaller WISP / ISP at 9× EBITDA — drives revenue + footprint + customer cross-sell + scale

Specificity rules (anti-hallucination)

- Every named federal program must be real (NTIA BEAD, USDA ReConnect, USDA Rural Utilities Service RUS, FCC Universal Service Fund USF, RDOF, CAF II, ECF E-Rate)
- Every named OSS / BSS must be real (Sonar, ITRoam, Powercode, Sandvine, Calix Cloud, RouteCo, Splynx, AzotelOSS)
- Every named equipment vendor must be real (Calix, Adtran, Nokia, Tibit Communications, Cambium, Mimosa, Tarana Wireless, Ubiquiti, Cisco, Juniper, Arista)
- Every named wholesale carrier must be real (Crown Castle, Zayo, Lumen, Windstream Wholesale, Frontier Communications, GTT Communications, AT&T; Wholesale, Verizon Wholesale)
- Every named association must be real (NTCA — Rural Broadband Association, WISPA — Wireless ISP Association, FBA — Fiber Broadband Association, ITTA, USTelecom)

Cost scale calibration

Telecom + ISP spend bands operate at network + equipment + per-passing scale:

Band	Spend type
1 — Under \$50,000	OSS module add, marketing, training
2 — \$50,000 to \$500,000	Single-tower FW deployment, OSS upgrade, sales team
3 — \$500,000 to \$5M	Small fiber buildout (1,000-5,000 passings), tower expansion
4 — \$5M to \$50M	Major FTTH / FW buildout, M&A; small ISP, OSS migration
5 — Over \$50M	Multi-county FTTH, regional expansion, large M&A;, BEAD-funded buildout

Persona schema

B2C (residential) + B2B (business + enterprise)

- Customer type (residential, SMB, enterprise, wholesale)
- Speed tier (entry / mid / gigabit / multi-gigabit)
- Decision driver (speed, reliability, price, customer service, local-vs-national)
- Acquisition channel (door-to-door, online, referral, named promo, wholesale partner)

- Multi-service bundle (internet only / + voice / + IPTV)
- LTV (monthly ARPU × retention × cross-sell)

Cycle / calendar model

Telecom + ISP runs on construction season + grant cycles + tech refresh:

Window	Trigger	Action
Construction season (Apr-Oct in north)	Network buildout window	Buildout capacity max + permit + locate work
Federal + state grant application cycles	BEAD, ReConnect, state programs	Grant prep + application discipline
Annual: customer renewal + churn cycle	Subscriber retention discipline	Retention motion + price-protect program
Annual: USF + FCC compliance	Regulatory compliance + USF funding	USF audit + Form 477 reporting
Tech refresh cycle (5-10 yr)	Equipment lifecycle	Capex planning + named equipment partners
Quarterly: financial + operational review	Operations health	Operational discipline + capacity planning
Daily / weekly: NOC + network ops	Network health	NOC discipline + outage management

Issue library

- Network reliability + outage handling — universal — frame as redundancy + named NOC platform
- Customer churn vs incumbent + new entrants — universal — frame as customer experience + bundle + price
- Capex burden + payback period — universal — frame as named grants + take-rate discipline
- Tower / pole / right-of-way access — universal — frame as named partnerships + ROW management
- OSS / BSS legacy + integration — universal — frame as named-platform modernization
- Customer service quality at scale — universal — frame as named platform + self-service portal
- Price competition from incumbents — universal — frame as customer experience + premium positioning
- Compliance burden (USF, CAF, BEAD) — universal — frame as compliance discipline + named platforms

Report sections

Section	Contents
Operator snapshot	Footprint, passings, subscribers, services mix, annual revenue, named comp set
Top 5 buildout + scale moves	Drawn from 13-category playbook
Federal + state grant pipeline	Named programs + application roadmap

Section	Contents
Network buildout + tech roadmap	FTTH / FW / hybrid + named equipment partners
B2B + wholesale + multi-service plan	Business + UCaaS + IPTV + wholesale opportunities
OSS / BSS + customer experience roadmap	Named platforms + investment priorities
Customer + named-account pipeline	Named B2B prospects + RFP cadence
Known industry challenges	Collapsed drawer

KPIs to track

- Subscribers + passings + take rate %
- ARPU by service tier
- Monthly churn % (target < 1.5% residential)
- Net Promoter Score
- Network availability % (target > 99.95%)
- Capex per passing + payback period
- EBITDA margin % (target > 35% mature)
- Grant funding capture rate

information.broadcasting_production — Broadcasting + production (radio + TV stations, video production cos, podcast networks, post-production)

Business model	B2B_advertising + B2B_service hybrid
NAICS examples	515112, 515120, 512110, 512199, 512240
Competitor radius	Local broadcast: market-defined (Nielsen DMA / Arbitron PPM-defined). National + syndicated: national. Production cos: regional + national + sometimes global on named project. Podcast networks: global within language.

Data sources

Source	Returns	Cost	Fallback
Nielsen + Comscore + Edison Research (audio)	Audience measurement + ratings	Subscription	Public summary
NAB (National Association of Broadcasters)	Industry benchmarks + named peers	Member-only	Public
BIA Advisory + RAB + TVB	Local broadcast revenue benchmarks	Subscription	Public

Source	Returns	Cost	Fallback
IAB podcast revenue + Edison Infinite Dial	Audio + podcast benchmarks	Free / sub	Direct
Named CRM + traffic systems (WideOrbit, Marketron, ENPS, iNEWS, Avid MediaCentral)	Internal + traffic + production	Subscription	Direct
Named ad-rep firms (Katz Media Group, Cumulus, iHeart Networks)	Ad rep + national sales benchmarks	Direct	Industry
FCC + database	License + ownership + violation data	Free public	Direct
Production Hub + StaffMeUp + LinkedIn (production crew)	Crew + freelance market	Free / sub	Direct

Review categories tracked

- Audience size + engagement
- Production quality
- On-air talent + brand
- Ad inventory + sponsor experience
- Schedule consistency
- Distribution + reach
- Multi-platform / digital extension
- Community engagement

Opportunity playbook (13 categories)

Category	Research basis	Example with named entity
Digital audio + podcast network buildout	Named platforms + podcast economics	Branded podcast network + named hosts + sales-rep partnership — adds \$400-1.2M/yr revenue + extends station brand to streaming audience
Streaming + smart-speaker distribution	Named platforms (TuneIn, iHeartRadio, Audacy, Spotify, Apple Music)	TuneIn Premium + iHeartRadio listing + Spotify station — extends reach beyond DMA + drives digital ad revenue
Local digital + content marketing services	Named platforms + local market opportunity	Sell content + social + digital services to named local advertisers — drives \$14-48k/mo at 50% margin (vs traditional ad-only)
Branded content + custom production line	Named branded content economics + sponsors	Branded content division producing podcasts + video for named clients — adds \$14-48k/project + drives recurring sponsor relationships

Category	Research basis	Example with named entity
Multi-platform programmatic ad sales	Named platforms (FreeWheel, Triton, AdsWizz, Magnite)	Programmatic audio + display via named platforms — captures fill on otherwise-unsold inventory at premium \$/CPM vs remnant
Named sales rep + national ad agency relationships	Named ad rep firms + national ad agencies	Katz Media Group + named national ad agency partnerships — drives 22-38% of revenue from national / regional ad budgets
Production studio rental + post-production line	Named studio + post-production economics	Studio rental + post-production for named local clients — adds \$14-48k/mo at 60% margin (utilizes idle studio capacity)
Live event production + streaming	Named event-streaming platforms + sponsors	Live event production for named local sponsors + streaming distribution — drives \$14-48k/event + sponsor relationships
Voiceover + commercial production line	Named voiceover talent + commercial production economics	In-house commercial production + named voiceover talent — captures \$1,200-4,800/spot + drives station-direct ad revenue
Multi-station + cluster strategy	Named cluster economics + named M&A; advisors	Cluster acquisition (3-5 stations in single market) — drives sales-rep efficiency + production sharing + sponsor cross-sell
NAB + community + remote broadcast events	NAB community + remote broadcast economics	Remote broadcast + community sponsorship + named local events — drives \$14-32k/event + community brand
AI / production-tech adoption	Named AI tools (Descript, OpenAI Whisper, ElevenLabs, Adobe Podcast)	AI transcription + voice cleanup + named editing tools — drops post-production time 60% + scales content output
Subscription + membership + reader-supported revenue	Named platforms + community-supported broadcasting	Member-supported model + named platform — drives recurring revenue from highest-engaged listeners

Specificity rules (anti-hallucination)

- Every named ratings service must be real (Nielsen, Comscore, Edison Research, BIA Advisory, RAB, TVB)
- Every named platform must be real (WideOrbit, Marketron, ENPS, iNEWS, Avid MediaCentral, FreeWheel, Triton Digital, AdsWizz, Magnite)
- Every named distribution platform must be real (TuneIn, iHeartRadio, Audacy, Spotify, Apple Music, Apple Podcasts, Amazon Music)
- Every named ad rep firm must be real (Katz Media Group, Cumulus Media, iHeart Networks)
- Every named association must be real (NAB National Association of Broadcasters, RAB Radio Advertising Bureau, TVB Television Bureau, IAB Interactive Advertising Bureau)

Cost scale calibration

Broadcasting + production spend bands operate at facility + talent + tech scale:

Band	Spend type
1 — Under \$5,000	Software license, training, marketing campaign
2 — \$5,000 to \$50,000	Equipment refresh, content creation, sponsor outreach
3 — \$50,000 to \$500,000	Studio upgrade, talent hire, production line buildout
4 — \$500,000 to \$5M	Multi-station expansion, cluster acquisition, major equipment
5 — Over \$5M	Major M&A;, multi-market expansion, full facility build

Persona schema

B2B_advertiser + B2C_audience hybrid

- Audience demographic + day-part listening pattern
- Acquisition channel (over-the-air, streaming, podcast, smart speaker, social)
- Engagement tier (passive listener / regular / member / sponsor relationship)
- Advertiser type (local SMB / regional / national agency / political)
- Decision driver for advertiser (CPM, audience match, integration, brand)
- LTV (audience: lifetime; advertiser: annual × retention)

Cycle / calendar model

Broadcasting runs on rating periods + political + retail cycles:

Window	Trigger	Action
Q4 holiday peak (Oct-Dec)	Retail + political ad surge	Sales max + inventory discipline
Political season (Q3-Q4 even years)	Political ad surge	Political ad discipline + premium pricing
Spring book / sweeps periods	Rating periods	Programming + audience build for ratings
Annual: ad rate card + renewal	Sponsor + advertiser cycles	Rate card discipline + sponsor retention
Daily / weekly: programming cadence	Content production	Content discipline + on-air execution
Annual: FCC license + EEO + ownership	Regulatory compliance	Compliance discipline + audit prep
Quarterly: ratings + revenue review	Operational health check	Ratings + revenue analytics + adjustment

Issue library

- Audience erosion vs streaming + podcast — universal — frame as multi-platform extension

- Local ad revenue compression — universal — frame as digital + content services + branded content
- National ad rep dynamics — universal — frame as named partnerships + direct national pursuit
- Talent retention + on-air brands — universal — frame as comp + culture + career path
- Ratings methodology shifts — universal — frame as multi-source measurement + platform diversification
- Programmatic CPM erosion — universal — frame as direct + premium positioning + named ad ops
- Capital + equipment refresh costs — universal — frame as named M&A; + cluster strategy
- AI + automation impact on production — universal — frame as adoption strategy + augmentation

Report sections

Section	Contents
Operation snapshot	Stations / shows / outputs, audience, advertiser mix, annual revenue, named comp set
Top 5 monetization + audience moves	Drawn from 13-category playbook
Multi-platform + digital extension plan	Named platforms + distribution roadmap
Branded content + production line	Named clients + custom production opportunities
Sales + ad ops + named partnerships	Direct + programmatic + named rep firm strategy
Programming + talent + community	On-air talent + community engagement plan
Audience personas (4 + lost listener)	Casual / loyal / member / advertiser / churned
Known industry challenges	Collapsed drawer

KPIs to track

- Audience size + reach by daypart
- Ad revenue + revenue per audience unit
- Local vs national ad revenue mix
- Digital + multi-platform revenue %
- Sponsor retention + new-sponsor count
- Programming cost / content velocity
- EBITDA margin % (target > 25% mature)
- Audience engagement (calls, social, events)

Engineering guidance

Sector 51 is Phase 5 specialty — global radius + B2B-subscription + ad-driven economics that no other sector exercises. Build information.saas_software first if covering Phase 5 (highest LTV per customer + highest valuation multiples + most M&A; activity). Media + publishing (51.B) is Phase 5 (most accessible barrier-to-entry but lowest margin). Telecom + ISP (51.C) is Phase 5 specialty (highest capital intensity, federal grant-driven). Broadcasting (51.D) is Phase 5 long-tail (declining traditional radio + TV but cluster M&A; active).

CHAPTER 61 — EDUCATIONAL SERVICES

Tutoring + test prep, K-12 private schools + Montessori, coding bootcamps + online programs, trade + technical schools. Phase 5 specialty — academic-calendar-driven, regulated by named accreditors.

One-line truth

A K-12 private school runs on enrollment cycles + accreditation + named-board governance. A coding bootcamp competes nationally on ISA / income-share + named hiring partners. A tutoring service is closer to local services but with academic-calendar seasonality. A trade school is regulated by named accreditors + state boards + Title IV. v3 has zero awareness of accreditation, ISA models, or academic cycles.

NAICS code count: 17 Sub-clusters: 4

education.tutoring_test_prep — Tutoring + test prep (1-on-1, group, academic, SAT/ACT/GRE/LSAT, music + arts lessons)

Business model	B2C_appointment
NAICS examples	611691, 611699, 611610
Modern variants	Online Tutoring Platform, Premium College Counseling, Music Lessons Studio, Math Olympiad Prep
Competitor radius	In-person: 12-25 mi service radius. Online: national + global. Specialty subjects + competition prep: regional + national depending on prestige.

Data sources

Source	Returns	Cost	Fallback
Google Places + Reviews	Reviews + photos + hours	\$0.02	Yelp Fusion
Google + Care.com + Wyzant + Tutor.com	Online tutoring marketplace + sentiment	Free / sub	Direct
Local school district directories	Named target customer schools	Free public	Direct
Named CRM + scheduling (TutorBird, Mindbody, Acuity, Practice Space, TutorCruncher)	Internal benchmarks + scheduling	Subscription	Direct
NTP (National Tutoring Association) + tutoring industry publications	Industry benchmarks + named peers	Member-only	Public summary

Source	Returns	Cost	Fallback
College Board + ACT + ETS + LSAC	Test admin data + named-prep programs	Free / sub	Direct
Local PTA + parent groups + Facebook parent communities	Named referral channels	Free	Direct
Named school counselors + admissions consultants	Named referral partners	Free / member	Direct

Review categories tracked

- Tutor expertise + outcome (test scores, grades)
- Tutor-student rapport
- Schedule reliability + flexibility
- Pricing transparency + value
- Curriculum / approach quality
- Parent communication + reporting
- Make-up policy + cancellation
- Specialty / advanced subject capability

Opportunity playbook (14 categories)

Category	Research basis	Example with named entity
Niche specialty positioning (subject, level, demographic)	Named tutoring competitors + market gaps	AP STEM specialist + named-school district focus + Math Olympiad / USAMO prep — premium \$/hr + sustained referral pipeline
Online + hybrid offering	Named platforms (Wyzant, Tutor.com, Outschool, Varsity Tutors)	Online + hybrid model + named platform — captures national + remote students + drops geographic constraint + drops studio cost
Test prep specialty + named test publishers	Named tests + prep curricula + named program licensing (Princeton Review, Kaplan, Magoosh)	SAT + ACT specialty + named curriculum + score guarantee — premium \$/hr + parent referral cycle
Premium college counseling + admissions consulting	Named admissions consultants + IECA membership	IECA-affiliated college counselor + named target schools + premium fee \$4-12k+ per student × 14 students = \$56-168k revenue
CRM + scheduling + parent portal	Named platforms (TutorBird, Mindbody, TutorCruncher, Practice Space)	TutorCruncher or TutorBird (\$79-249/mo) + parent portal + automated billing — drops admin time 60% + drives parent satisfaction

Category	Research basis	Example with named entity
Group + cohort tutoring model	Group economics + named programs	Group cohort tutoring (4-6 students) + named curriculum — drives 2-4× hourly revenue + drops per-student cost for parents
Subject-matter expert + named-credential tutors	Named credentials + premium positioning	Named-university grad tutors + subject-matter expertise — captures premium \$/hr (\$120-240/hr) vs commodity tutoring (\$45-65/hr)
Sales + funnel + content engine	Named platforms (HubSpot, ConvertKit) + tutoring content	Topic-cluster content (How to Prep for SAT, AP Calculus tips) + email funnel — drives 8-14 qualified inquiries/mo at \$0 ad spend after 12 mo
Named-school + named-PTA partnership	Named school + parent organization channels	Named local school + PTA partnership — drives 22-38% of new students at \$0 CAC + community brand
Music / arts lessons specialty + studio brand	Named music + arts associations (NAfME, MTNA, Suzuki, Music Together)	Music Teachers National Association MTNA cert + Suzuki Method specialty + studio brand — captures premium \$/lesson + community presence
Specialty programs (gifted + talented, learning disabilities, ADHD)	Named specialty cert + programs (Wilson Reading System, Orton-Gillingham, Lindamood-Bell)	Orton-Gillingham + Wilson Reading + dyslexia specialty — captures premium \$/hr from named-need families + low competition
Subscription / package / membership pricing	Named subscription pricing + tutoring economics	Monthly membership \$480/mo (8 sessions) — drives recurring revenue + retention + drops per-session sales friction
Acquisition + tutoring center roll-up	Named M&A; + tutoring center economics	Acquire smaller tutoring center / studio — drives student book + tutor team + footprint expansion
Tutor recruiting + retention + comp model	Named recruiting platforms + tutor retention	Tutor recruiting via LinkedIn + university job boards + named compensation model — drives tutor quality + retention + premium positioning

Specificity rules (anti-hallucination)

- Every named platform must be real (TutorBird, TutorCruncher, Mindbody, Acuity, Practice Space, Wyzant, Tutor.com, Varsity Tutors, Outschool)
- Every named test must be real (SAT, ACT, GRE, GMAT, LSAT, MCAT, AP exams, IB, SSAT, ISEE)
- Every named curriculum must be real (Princeton Review, Kaplan Test Prep, Magoosh, Khan Academy, Eureka Math, IXL)
- Every named association must be real (NTA National Tutoring Association, IECA Independent Educational Consultants Association, NAfME, MTNA, Suzuki, NCEE)
- Every named specialty cert must be real (Orton-Gillingham, Wilson Reading System, Lindamood-Bell, Slingerland, MTNA cert)

Cost scale calibration

Tutoring + test prep spend bands operate at small-business + per-tutor scale:

Band	Spend type
1 — Under \$500	Platform setup, partnership outreach, training
2 — \$500 to \$5,000	CRM + scheduling software, marketing, signage
3 — \$5,000 to \$50,000	Studio space, tutor hire, curriculum licensing
4 — \$50,000 to \$500,000	Multi-tutor expansion, second location, technology
5 — Over \$500,000	Multi-location expansion, M&A; acquisition

Persona schema

B2B_buyer (parent) + B2C_user (student)

- Parent decision driver (results, credentials, schedule, price, fit)
- Student level + subject focus + goal
- Acquisition channel (Google, school referral, neighbor, named PTA)
- Engagement type (one-off / weekly / cohort / program)
- Outcome focus (grade improvement / test prep / advanced / specialty)
- LTV (sessions × duration × multi-child × referral)

Cycle / calendar model

Tutoring + test prep run on academic calendar + test cycle + admissions:

Window	Trigger	Action
Back-to-school (Aug-Sep)	New academic year + tutor demand surge	Capacity max + new-student onboarding
SAT / ACT prep cycles	Test admin dates throughout year	Test prep cohort discipline + named curriculum
Q4 college admissions push (Sep-Jan)	Application + essay + early decision	College counseling capacity max
AP exam season (Apr-May)	AP exam prep peak	AP-prep capacity + group cohorts
Summer (Jun-Aug)	Summer prep + camps + advanced work	Summer program + intensive offerings
Daily: scheduling discipline	Operational	Scheduling discipline + tutor capacity
Annual: tutor recruiting + onboarding	Tutor team building	Recruiting + named-credential pipeline

Issue library

- Tutor turnover + retention — universal — frame as comp + culture + named programs

- Schedule reliability + cancellation — universal — frame as cancellation policy + named platform
- Outcome measurement + score guarantees — universal — frame as named-curriculum + outcome tracking
- Online vs in-person mix — universal — frame as hybrid model + named platforms
- Sales-funnel quality + parent decision-making — universal — frame as content + nurture + consultation
- Pricing pressure from commodity tutoring — universal — frame as specialty + named-credential positioning
- Multi-child family discount discipline — universal — frame as transparent pricing + family discount tier
- Student outcome variability — universal — frame as named curriculum + tutor selection discipline

Report sections

Section	Contents
Center snapshot	Tutors, students, subject mix, annual revenue, named comp set
Top 5 growth + retention moves	Drawn from 14-category playbook
Specialty / niche positioning roadmap	Subject + demographic + outcome specialty
Tutor recruiting + retention plan	Named credentials + comp + culture
Tech + CRM + parent portal roadmap	Named platforms + investment priorities
Customer + named-school pipeline	Named schools + PTA + counselor referrals
Customer personas (4 + lost student)	Subject / test prep / specialty / churned
Known industry challenges	Collapsed drawer

KPIs to track

- Annual revenue + revenue per tutor
- Active student count + retention
- Avg sessions per student
- Student outcome (test score lift, grade improvement)
- Tutor retention 12-mo
- Customer acquisition cost by channel
- Avg Google rating (target > 4.7)
- Referral % of new students (target > 35%)

education.k12_private — K-12 private + Montessori + parochial schools, charter schools, religious schools

Business model B2C_subscription (long-term)

NAICS examples	611110
Modern variants	Montessori, Waldorf, Catholic / Christian, College Prep, Special Needs
Competitor radius	Local: 15-30 mi commute radius. Specialty / boarding: regional + national. Religious / parochial: defined by named denomination network.

Data sources

Source	Returns	Cost	Fallback
NAIS (National Association of Independent Schools)	Industry benchmarks + named peers	Member-only	Public summary
Niche.com + GreatSchools.org + Private School Review	School ratings + reviews	Free	Direct
Named accreditors (NAIS, NEASC, MSA, SACS, AdvancED, AMS Montessori, AWSNA Waldorf)	Accreditation status + named peers	Member-only	Public
ISACS + state independent schools association	Regional benchmarks	Member-only	Public
Named SIS + LMS (Veracross, Blackbaud, FACTS, Finalsight, Ravenna)	Internal + tech benchmarks	Subscription	Direct
ERB Educational Records Bureau + ISEE / SSAT prep	Admissions test data	Member-only	Public
Named admissions software (Ravenna, FinalsightEnroll, Senior Systems, Veracross Admissions)	Admissions tech + benchmarks	Subscription	Direct
Local + diocesan + religious denominational data	Religious school networks	Free / member	Direct

Review categories tracked

- Academic outcomes (test scores, college admissions)
- Faculty quality + retention
- Class size + student-teacher ratio
- Curriculum + program (Montessori, Waldorf, IB, AP)
- Community + culture + values
- Tuition value + financial aid
- Facilities + extracurriculars
- Communication + parent engagement

Opportunity playbook (13 categories)

Category	Research basis	Example with named entity
Accreditation + named-accreditor cycle	Named accreditors (NAIS, NEASC, MSA, SACS, AdvancED, NAEYC, AMS Montessori, AWSNA Waldorf, NCEA Catholic)	NAIS member + regional accreditation + AMS Montessori cert (where applicable) — drives parent trust + premium tuition + admissions appeal
Admissions funnel + technology	Named admissions platforms (Ravenna, Finalsight, Senior Systems, Veracross Admissions, Blackbaud Admissions)	Ravenna platform + funnel discipline + named-school events — drives 22-38% application growth + drops admissions cost per family
Tuition + financial aid optimization	Named platforms (FACTS Grant + Aid, TADS, SSS by NAIS)	FACTS or SSS need-based aid + indexed tuition — captures middle-income families + drives diversity + drops tuition discount discipline
STEM / specialty program addition (IB, robotics, languages, music conservatory)	Named specialty programs (IB Diploma, FIRST Robotics, FLL, AP, named-language immersion)	IB Diploma program + named language immersion (Mandarin, French, Spanish) — captures premium tuition + admissions appeal + retention
Religious + values-based community engagement	Named denominational networks + community programs	Catholic / Lutheran / Jewish / Christian / Islamic school + named diocesan or denominational program — drives community + retention + tuition support
Faculty retention + comp + named-cert	NAIS faculty comp benchmarks + named cert	Faculty retention program + comp benchmark to NAIS percentile + named-cert support — drives faculty quality + parent trust
Capital campaign + endowment + named-donor program	Named campaign consultants + advancement benchmarks	Capital campaign with named consultant + lead donor identification — raises \$4-40M+ for facilities + endowment + financial aid
Marketing + named-school events + open houses	Named school marketing + admissions cycles	Open house + admitted-student events + named alumni speakers — drives 28-48% admissions yield improvement
Tech stack + LMS + parent portal	Named platforms (Veracross, Blackbaud, FACTS SIS, Finalsight, Schoology, Canvas, Google Classroom)	Veracross or Blackbaud SIS + LMS + parent portal — drives parent satisfaction + drops admin burden + drives retention

Category	Research basis	Example with named entity
Diversity + equity + inclusion + scholarship	Named DEI programs + scholarship programs (Steppingstone, Prep for Prep, A Better Chance)	Named scholarship partnership (Prep for Prep, A Better Chance) — drives community + diversity + drives admissions appeal
Boarding + extended-day + summer programs	Named boarding + extended programs	Boarding (where named markets support) or extended-day + summer programs — drives revenue + utilization + retention
College counseling + named-college outcomes	Named college counseling + IECA + outcomes tracking	In-house college counseling + named-college outcomes (Ivy + Top 50) — drives admissions appeal + premium tuition justification
Strategic plan + named-board governance	Named board governance + NAIS strategic plan benchmarks	Named-board strategic plan + 5-yr capital + program roadmap — drives institutional discipline + funder confidence

Specificity rules (anti-hallucination)

- Every named accreditor must be real (NAIS, NEASC New England, MSA Middle States, SACS Southern, AdvancED, NAEYC, AMS American Montessori Society, AWSNA Association of Waldorf Schools North America, NCEA National Catholic Educational Association, ACSI Association of Christian Schools International)
- Every named SIS / LMS must be real (Veracross, Blackbaud, FACTS, Finalsight, Ravenna, Senior Systems, Schoology, Canvas, Google Classroom, Brightspace)
- Every named curriculum must be real (IB International Baccalaureate, AP Advanced Placement, Montessori, Waldorf, Reggio Emilia, FIRST Robotics, FLL FIRST LEGO League)
- Every named scholarship program must be real (Steppingstone, Prep for Prep, A Better Chance, Black Student Fund)
- Every named association must be real (NAIS, ISACS Independent Schools Association of the Central States, AISGW, ISBOA business officers)

Cost scale calibration

K-12 private school spend bands operate at school + per-student + capital scale:

Band	Spend type
1 — Under \$5,000	Software module, training, marketing template
2 — \$5,000 to \$50,000	Marketing campaign, faculty cert, single-program addition
3 — \$50,000 to \$500,000	SIS + LMS upgrade, faculty hire, single-grade expansion
4 — \$500,000 to \$5M	Major program addition, facility renovation, capital campaign launch
5 — Over \$5M	New facility, major capital campaign, multi-campus expansion

Persona schema

B2C (parent + family) + B2C (student)

- Family decision driver (academic, religious, values, community, outcomes)
- Family income tier + financial aid need
- Student level + special needs
- Acquisition channel (community, alumni, named open house, marketing)
- Multi-child family + sibling discount
- LTV (tuition × years × siblings × alumni giving)

Cycle / calendar model

K-12 schools run on academic calendar + admissions cycle + capital + accreditation cycles:

Window	Trigger	Action
Admissions season (Sep-Mar)	Application + decision + enrollment cycle	Admissions discipline + open houses + named events
Re-enrollment cycle (Jan-Mar)	Family re-enrollment decision	Retention motion + tuition + financial aid discipline
Annual: tuition setting + financial aid	Annual tuition decision	Tuition discipline + named-platform aid
Summer (Jun-Aug)	Faculty hire + program planning + camps	Summer programs + faculty onboarding
Annual: capital campaign + giving cycle	Annual fund + named-donor cycles	Advancement discipline + named-donor cultivation
Multi-year: accreditation cycle (5-10 yr)	Accreditation visit prep	Accreditation discipline + self-study
Quarterly: board + committee + governance	Board discipline	Governance + strategic plan execution

Issue library

- Enrollment + retention pressure — universal — frame as admissions + retention motion
- Tuition affordability + financial aid burden — universal — frame as named-platform aid + endowment
- Faculty retention + compensation — universal — frame as comp + culture + named cert support
- Tech / SIS / LMS modernization — universal — frame as named platforms + integration
- Capital + facility + endowment needs — universal — frame as named campaign + advancement
- Marketing + admissions yield — universal — frame as named events + community + alumni
- Diversity + equity + inclusion — universal — frame as named scholarship + intentional community
- Accreditation + governance discipline — universal — frame as continuous improvement + strategic plan

Report sections

Section	Contents
School snapshot	Enrollment, faculty, programs, tuition, named comp set

Section	Contents
Top 5 enrollment + retention moves	Drawn from 13-category playbook
Admissions + marketing roadmap	Named platforms + funnel + events strategy
Faculty + program roadmap	Faculty hire + named programs + cert pursuit
Tuition + financial aid + capital plan	Named platforms + advancement strategy
Tech stack + parent portal roadmap	Named SIS / LMS + investment priorities
Family personas (4 + lost family)	Academic / religious / values / community / churned
Known industry challenges	Collapsed drawer

KPIs to track

- Enrollment + retention rate (target > 92%)
- Application + admit + yield rate
- Faculty retention 12-mo (target > 92%)
- Tuition + financial aid discount rate
- College placement outcomes (where applicable)
- Annual fund + capital giving
- Accreditation status + cycle position
- Net tuition revenue per student

education.bootcamps_online — Coding bootcamps + online programs (full-stack, data science, UX, cybersecurity, professional certificates)

Business model	B2C_subscription + B2B_subscription hybrid
NAICS examples	611420, 611430, 611699
Modern variants	ISA-Funded Bootcamp, B2B Tech Training, MOOC Aggregator, Self-Paced Online Program
Competitor radius	Online: global. In-person bootcamp: 50-200 mi. Hybrid: defined by named cohort city. Niche / specialty: global within language.

Data sources

Source	Returns	Cost	Fallback
Course Report + SwitchUp + bootcamp review aggregators	Industry rankings + outcomes	Free / sub	Direct

Source	Returns	Cost	Fallback
CIRR (Council on Integrity in Results Reporting)	Outcomes audit + named participating bootcamps	Free public	Direct
Named platforms (Coursera, edX, Udemy, Pluralsight, LinkedIn Learning, Skillshare, MasterClass)	Online learning benchmarks	Public / direct	Direct
Named hiring partners + alumni networks	Named employer relationships	Direct	Industry
LinkedIn + Glassdoor + Indeed	Alumni outcomes + job placement intel	Free / sub	Direct
ApplicationLab + Career Karma + named bootcamp aggregator	Lead-gen + comparison platforms	Free / sub	Direct
Named LMS + course platforms (Teachable, Thinkific, Kajabi, LearnDash, Podia, Mighty Networks)	Internal benchmarks + tools	Subscription	Direct
Stripe + Affirm + named ISA / financing partners	Financing + ISA economics	Direct	Industry

Review categories tracked

- Job placement rate + named hiring outcomes
- Curriculum quality + currency
- Instructor expertise + engagement
- Career services + named hiring partnerships
- Cohort experience + community
- Pricing transparency + ISA terms
- Refund + outcome guarantees
- Time-to-job + outcome reporting

Opportunity playbook (13 categories)

Category	Research basis	Example with named entity
CIRR membership + outcomes transparency	CIRR Council on Integrity in Results Reporting + named members	CIRR membership + audited outcomes reporting — drives 38% applicant trust lift + premium positioning vs unverified competitors
Income Share Agreement (ISA) + outcomes-based financing	Named ISA platforms (Leif Capital, Vemo Education, Avenify, Edly)	ISA model via Leif Capital + outcome-based pricing — drives 38% application growth + aligns student-bootcamp incentives + drops upfront barrier

Category	Research basis	Example with named entity
Named hiring partner + corporate-direct pipeline	Named tech employer hiring partnerships	Hiring partnership with named tech employers + B2B-direct pipeline — drives 60-80% of placement at \$0 student-side cost
B2B + corporate training line	B2B training market + named platforms (Pluralsight Skills, Coursera Business, Udemy Business, A Cloud Guru)	B2B corporate training contracts — adds \$480k-2.4M/yr revenue at 60% margin (vs B2C 22% margin) + named employer relationships
Specialty curriculum (cybersecurity, AI / ML, data engineering)	Named specialty markets + premium pricing	Cybersecurity bootcamp + named-cert prep (CompTIA, CISSP) — captures growing market segment + premium tuition + low competition
Online + cohort-based + community model	Named cohort-based course platforms (Maven, Reforge, Section)	Cohort-based model + named community platform + premium pricing — drives community + retention + premium positioning
Career services + outcomes engine	Named career-services frameworks	Dedicated career services team + named hiring funnel + alumni network — drives placement + outcomes + drives admissions appeal
Marketing + content + community engine	Named platforms + content economics	YouTube + LinkedIn + content engine + named instructors + alumni stories — drives 14-28k organic monthly visitors + applicant pipeline
Acquisition / consolidation strategy	Named M&A; + valuation multiples (3-6× revenue)	Acquire complementary bootcamp / online program — drives revenue + curriculum + named hiring partner expansion
Self-paced + on-demand + multi-format	Named platforms + content economics	Self-paced + on-demand + named platform partnership (Coursera, edX) — drives passive revenue at low cost-to-deliver
University / accredited partnership	Named university partnerships + accredited credit	University partnership + accredited credit pathway — drives credibility + Title IV access + premium positioning
AI / automation in curriculum + delivery	Named AI tools (ChatGPT, Claude, named coding assistants)	AI-augmented coding + project review — drops instructor cost + scales personalization + drives outcomes
Diversity + scholarship + named-program partnerships	Named scholarship programs + diversity initiatives	Named scholarship + diversity program (Code2040, Adobe Digital Academy partnership) — drives community + diversity + funder partnerships

Specificity rules (anti-hallucination)

- Every named CIRR-participating bootcamp / outcomes auditor must be real (CIRR, BPPE California, named state regulatory bodies)
- Every named ISA / financing platform must be real (Leif Capital, Vemo Education, Avenify, Edly, Climb Credit, Affirm)
- Every named platform must be real (Coursera, edX, Udemy, Pluralsight, LinkedIn Learning, Skillshare, MasterClass, Teachable, Thinkific, Kajabi, Podia, LearnDash, Mighty Networks, Maven, Reforge, Section)
- Every named cert program must be real (CompTIA Security+/Net+/A+, CISSP, AWS Certified, Google Cloud, Microsoft Azure, Cisco, named tech-vendor certs)
- Every named bootcamp aggregator must be real (Course Report, SwitchUp, Career Karma, ApplicationLab)

Cost scale calibration

Bootcamp + online program spend bands operate at curriculum + GTM + tech scale:

Band	Spend type
1 — Under \$10,000	Platform license, single course launch, marketing
2 — \$10,000 to \$100,000	Curriculum buildout, content team, marketing campaign
3 — \$100,000 to \$1M	Multi-program expansion, named hiring partnerships, B2B sales
4 — \$1M to \$10M	Multi-cohort scale, geographic expansion, M&A; small
5 — Over \$10M	Major M&A;,, multi-vertical expansion, university partnership

Persona schema

B2C (career-changer) + B2B (corporate buyer) hybrid

- Student type (career changer, upskiller, recent grad, corporate-sponsored)
- Decision driver (placement rate, outcomes, price, ISA, community, time-to-job)
- Acquisition channel (named aggregator, content, social, alumni, employer)
- Outcome focus (job placement / promotion / specialty / hobby)
- Financing model preference (upfront / loan / ISA / employer-paid)
- LTV (tuition × cohort × upsell × alumni referral)

Cycle / calendar model

Bootcamps + online programs run on cohort + admissions + employer hiring cycles:

Window	Trigger	Action
Cohort cadence (typically 6-12 wks)	Continuous cohort model	Cohort discipline + admissions funnel
Annual: corporate L&D; budget cycle (Q4 → Q1)	B2B budget approval	B2B sales discipline + named-account targeting
Hiring cycles (Q1 + Q3 strongest)	Tech hiring peaks	Career services intensity + employer outreach

Window	Trigger	Action
Quarterly: outcomes audit + CIRR reporting	Outcomes discipline	Outcomes tracking + transparent reporting
Annual: curriculum refresh	Tech stack evolution	Curriculum discipline + named-tech currency
Daily / weekly: instruction + content cadence	Operational	Content + instruction discipline
Annual: hiring partnership renewal	Named partner relationships	Partnership discipline + cross-cohort placement

Issue library

- Job placement rate vs marketing claims — universal — frame as CIRR + outcomes audit + transparency
- Curriculum currency vs tech stack churn — universal — frame as continuous refresh + named-tech
- Instructor recruitment + retention — universal — frame as comp + culture + alumni instructor pipeline
- ISA + financing economics + risk — universal — frame as named-platform partner + cohort discipline
- B2B vs B2C revenue mix — universal — frame as B2B GTM motion + named-account targeting
- Online vs in-person vs hybrid — universal — frame as cohort-based + community model
- Marketing + admission funnel ROI — universal — frame as content + named-aggregator discipline
- Outcomes-based regulation (state BPPE, federal) — universal — frame as compliance + transparency

Report sections

Section	Contents
Program snapshot	Cohorts, students, programs, hiring partners, annual revenue, named comp set
Top 5 strategic moves	Drawn from 13-category playbook
Outcomes + CIRR + transparency roadmap	Named programs + audit pursuit
Hiring partnership + B2B pipeline	Named employer + corporate-direct opportunities
Specialty / vertical curriculum roadmap	Named specialty + named cert pathways
GTM + content + community engine	Named platforms + content strategy
Capital + financing + ISA strategy	Named partners + financing model
Known industry challenges	Collapsed drawer

KPIs to track

- Annual revenue + revenue per cohort
- Job placement rate (target > 80% within 6 mo)
- CIRR-audited outcomes

- Time-to-job (target < 6 mo)
- Cohort completion rate (target > 80%)
- B2B / B2C revenue mix
- Customer acquisition cost by channel
- Net Promoter Score from grads

education.trade_technical — Trade + technical schools (welding, HVAC, electrical, automotive, cosmetology, CDL truck driving, healthcare aide)

Business model	B2C_subscription (short-term) + Title IV
NAICS examples	611513, 611519
Competitor radius	Local: 30-100 mi commute radius. Specialty / niche programs: regional 200+ mi. CDL: defined by state regulations.

Data sources

Source	Returns	Cost	Fallback
ACCSC + ACCET + COE accreditation databases	Named accreditors + member schools	Free public	Direct
Federal Student Aid + Title IV school code database	Title IV eligibility + named-school data	Free public	Direct
Career One Stop + state career center databases	Job market + named industry employers	Free	Direct
Named industry licensing boards (state HVAC, electrical, cosmetology, CDL, nursing aide)	License pass rate + named-school data	Free public	Direct
NCCER + named industry training programs	Industry curriculum + cert benchmarks	Member-only	Public
Named SIS + Title IV compliance (Campus Cafe, Anthology, FAME, FA Solutions)	Internal + Title IV compliance	Subscription	Direct
IBHE + state postsecondary boards	State-level oversight + named schools	Free public	Direct
Named employers + industry hiring partnerships	Named employer relationships	Direct	Industry

Review categories tracked

- Job placement rate + named employer outcomes

- License pass rate + named-board outcomes
- Hands-on training + equipment quality
- Instructor expertise (named industry cert)
- Financial aid + Title IV discipline
- Job placement + career services
- Schedule flexibility (day / evening / weekend)
- Specialty / advanced training

Opportunity playbook (14 categories)

Category	Research basis	Example with named entity
Accreditation maturation (ACCSC, ACCET, COE, named regional)	Named accreditors + Title IV requirements	ACCSC or ACCET accreditation + Title IV eligibility — captures federal financial aid + drives enrollment + premium tuition vs unaccredited
Title IV + Pell Grant + federal aid eligibility	Title IV requirements + named compliance	Title IV eligibility + federal aid processing — captures federal financial aid for 60-80% of students at named-school + drives enrollment
Named industry employer partnership + apprenticeship	Named employer partnerships + DOL Registered Apprenticeship	DOL Registered Apprenticeship + named employer + RAP partner — captures hiring pipeline + earn-while-you-learn + drives enrollment
Named industry licensing + cert prep	Named state licensing boards + cert prep	State HVAC + electrical + cosmetology license prep + named cert (NATE, EPA 608) — drives placement + premium tuition
High-demand specialty programs (CDL, healthcare aide, welding, HVAC)	Named labor market data + named programs	CDL Class A program + named carrier hiring partner — captures growing market + premium tuition + named hiring pipeline
WIOA + workforce development funding	Named WIOA programs + state workforce boards	WIOA Title I + state workforce funding partnership — captures public-funded students + drives revenue at modest discount
Career services + placement + named employers	Named career-services frameworks + outcomes tracking	Dedicated career services + named employer hiring fairs + placement guarantees — drives outcomes + drives admissions appeal
SIS + Title IV compliance modernization	Named platforms (Campus Cafe, Anthology Student, FAME, FA Solutions, Powerschool Unified Insights, Jenzabar)	Campus Cafe or Anthology + Title IV processing + automated compliance — drops compliance cost + drives audit pass rate
Veterans + GI Bill eligibility	VA approval + named GI Bill programs	VA approval + named GI Bill processing + Yellow Ribbon participation — captures veteran students + premium tuition support

Category	Research basis	Example with named entity
Healthcare specialty (nursing aide, medical assistant, dental hygiene)	Named healthcare licensing + outcomes	CNA + medical assistant + named cert (NHA, AMT) prep — premium tuition + low competition + named healthcare employer pipeline
Hybrid + online + flexible scheduling	Named LMS + hybrid delivery	Hybrid + evening + weekend scheduling — captures working adult demographic + drives enrollment from full-time workers
Marketing + content + community engagement	Named platforms + community outreach	Named-employer testimonial videos + named-industry presence + community visibility — drives 14-22 qualified inquiries/mo at moderate CAC
Multi-program + multi-campus expansion	Named expansion economics + accreditation portability	Add second campus or new program — drives revenue + capacity + named employer expansion
Strategic acquisition / consolidation	Named M&A; + valuation (3-6× EBITDA + accreditation premium)	Acquire smaller trade / technical school — drives revenue + accreditation expansion + named employer + program diversification

Specificity rules (anti-hallucination)

- Every named accreditor must be real (ACCSC Accrediting Commission of Career Schools, ACCET Accrediting Council for Continuing Education and Training, COE Council on Occupational Education, named regional accreditors)
- Every named federal program must be real (Title IV, Pell Grant, WIOA Workforce Innovation and Opportunity Act, GI Bill, Yellow Ribbon, DOL Registered Apprenticeship, RAP)
- Every named industry cert must be real (NATE HVAC, EPA 608, ASE auto, AWS welding, CDL Class A/B, NHA medical assistant, AMT, named state cosmetology + electrical + plumbing)
- Every named SIS must be real (Campus Cafe, Anthology Student, FAME, FA Solutions, Jenzabar, Powerschool Unified Insights)
- Every named association must be real (CECU Career Education Colleges and Universities, NCCER, named industry training programs)

Cost scale calibration

Trade + technical school spend bands operate at facility + equipment + per-program scale:

Band	Spend type
1 — Under \$10,000	Software + cert application + marketing
2 — \$10,000 to \$100,000	Equipment + named cert prep + curriculum
3 — \$100,000 to \$1M	Program addition + accreditation + facility
4 — \$1M to \$10M	Multi-program expansion + second campus + accreditation expansion
5 — Over \$10M	Major M&A; + multi-state expansion + new facility + accreditation transfer

Persona schema

B2C (career-changer + working adult + recent HS grad)

- Student type (recent HS grad / career changer / displaced worker / veteran / working adult)
- Decision driver (placement rate, license pass rate, financial aid, schedule, named-employer)
- Financing source (Title IV / WIOA / GI Bill / employer / private)
- Acquisition channel (named-employer / state workforce / referral / Google / community)
- Schedule constraint (full-time / evening / weekend / hybrid)
- LTV (tuition × cohort × multi-program × alumni referral)

Cycle / calendar model

Trade + technical schools run on cohort + admissions + Title IV + accreditation cycles:

Window	Trigger	Action
Cohort start cycles (multiple per year)	Multiple program starts	Cohort discipline + admissions discipline
Federal student aid (FAFSA) cycle (Oct-March)	Federal aid processing	Title IV processing + financial aid discipline
WIOA + state workforce funding cycle	Public funding cycles	WIOA partnerships + state-level discipline
Annual: accreditation reporting + Title IV compliance	Accreditation + federal compliance	Pre-audit prep + management review
Quarterly: outcomes + placement reporting	Outcomes discipline	Outcomes tracking + named-employer placement
Annual: industry-cert pass rate audit	Named-board pass rates	License pass rate discipline
Daily / weekly: instructional + lab cadence	Operational	Hands-on training discipline

Issue library

- Title IV compliance + audit risk — universal — frame as named platform + compliance discipline
- Job placement rate + outcomes pressure — universal — frame as career services + named-employer partnerships
- License pass rate + cert outcomes — universal — frame as named curriculum + cert prep
- Accreditation maintenance + risk — universal — frame as continuous improvement + self-study
- Financial aid processing + family experience — universal — frame as named platform + aid counselor
- Marketing + admission funnel ROI — universal — frame as named-employer + community
- Equipment + facility refresh costs — universal — frame as named-employer partnership + grant funding
- Workforce funding source diversification — universal — frame as Title IV + WIOA + named-employer mix

Report sections

Section	Contents
School snapshot	Programs, students, instructors, named-employer partners, annual revenue, named comp set
Top 5 strategic moves	Drawn from 14-category playbook
Accreditation + Title IV + cert roadmap	Named programs + compliance pursuit
Named employer + apprenticeship pipeline	Named partner + RAP + hiring relationships
Program addition + specialty roadmap	Named programs + cert + license pathways
Funding source + financial aid plan	Title IV + WIOA + GI Bill + employer-paid mix
Tech stack + SIS + compliance roadmap	Named platforms + investment priorities
Known industry challenges	Collapsed drawer

KPIs to track

- Annual revenue + revenue per program
- Enrollment + retention + completion rate
- Job placement rate (target > 70% within 6 mo)
- License pass rate (named industry boards)
- Title IV / federal aid pass rate (target 100%)
- Cohort retention
- Named-employer placement rate
- Customer acquisition cost by channel

Engineering guidance

Sector 61 is Phase 5 specialty — academic-calendar-driven + accreditation-regulated. Build `education.tutoring_test_prep` first if covering Phase 5 (most accessible barrier-to-entry + most local). K-12 private (61.B) is Phase 5 (highest LTV per family + accreditation-driven). Bootcamps (61.C) is Phase 5 (most M&A; active + ISA + outcomes-driven). Trade + technical (61.D) is Phase 5 (most regulated + Title IV-driven + named industry partnerships).

OUT-OF-SCOPE SECTORS

Sectors 55 + 92 — out of BizRadar scope

BizRadar v4 covers operating businesses with retail-buyer relevance — entities with storefronts (physical or digital), customers (named accounts or anonymous consumers), competitors, and a meaningful improvement opportunity surface. Sectors 55 and 92 don't qualify. This out-of-scope classification happens at Master Router (Layer 0.5) before any data fetching, so there's zero cost incurred on out-of-scope inputs.

Sector 55 — Management of Companies and Enterprises

NAICS-4 code count 3

Retail-buyer relevance Effectively zero retail-buyer relevance

Rationale for exclusion

Sector 55 covers holding companies, corporate offices, and enterprise management — entities that exist on paper for tax, governance, and legal-structure purposes. They are not businesses that compete for customers in any meaningful sense. A holding company has no storefront, no review profile, no operating customers, no marketing function, and no competitor radius. The operating businesses owned by the holding company are classified in their own NAICS sectors (e.g. a holding company that owns a hotel is a 55, but the hotel itself is a 72). BizRadar v4 covers the operating businesses, not their corporate parents.

Engineering action

Master Router classifies any NAICS-2 = 55 input as `out_of_scope` and returns a structured message: "BizRadar covers operating businesses, not holding companies. If you're looking for a report on a specific operating business owned by this holding company, search by that operating business name." No report is generated. Refund or no-charge handling on the payment layer.

Edge cases

- Family offices that operate as small holding cos with one or two operating businesses — route to the operating business profile if identifiable
- Real estate holding cos (LLCs holding properties) — route to `real_estate.*` profiles where there is operating activity
- Corporate HQ functions of large operating businesses — these still have an operating business at the parent level (e.g. Walmart corporate is still classified by Walmart's retail operations)

Sector 92 — Public Administration

NAICS-4 code count 29

Retail-buyer relevance Government entities, not businesses

Rationale for exclusion

Sector 92 covers federal, state, and local government entities — courts, legislative bodies, executive agencies, regulatory agencies, military, public safety, public finance. These are not businesses competing for customers and have no retail-buyer relevance. BizRadar v4 is a tool for business owners and operators; government entities are not within scope. Note that government contractors (e.g. an IT firm doing federal work) are classified under their own NAICS sector (e.g. 541512 IT consulting), not under 92.

Engineering action

Master Router classifies any NAICS-2 = 92 input as `out_of_scope` and returns a structured message: "BizRadar covers private-sector businesses. Government entities are not within scope." No report is generated. Refund or no-charge handling on the payment layer.

Edge cases

- Quasi-governmental entities (port authorities, water districts, transit agencies) — classified in 92 but may have meaningful operating-business characteristics. Out-of-scope by default; manual override possible if the entity has named-customer service operations.
- Government-owned utilities (municipal electric, water) — out-of-scope by default but adjacent to sector 22 utilities operating-business logic.
- Federal contractors — these are NOT 92; they're classified by their service line (e.g. 541512 IT consulting, 541330 engineering). Federal contractor angle is captured in those sectors' playbooks.

Master Router pseudocode

The out-of-scope short-circuit happens at Layer 0.5 before any data fetching, so there's zero cost incurred on out-of-scope inputs:

```
function masterRouter(naicsCode, businessTypeQuery) {
  const naics2 = naicsCode.substring(0, 2);

  // Out-of-scope short-circuit
  if (naics2 === "55") {
    return { profileId: null, status: "out_of_scope", reason: "holding_company",
      message: "BizRadar covers operating businesses, not holding companies." };
  }
  if (naics2 === "92") {
    return { profileId: null, status: "out_of_scope", reason: "public_administration",
      message: "BizRadar covers private-sector businesses." };
  }

  // Continue to in-scope routing for sectors 11 through 81
  return routeInScopeSector(naics2, naicsCode, businessTypeQuery);
}
```

PART 3 — NAICS-4 FAILURE-RATE DATABASE

Establishment survival by NAICS-4 industry

Database of 302 NAICS-4 codes covering all 18 in-scope sectors plus selected adjacent codes. Each entry shows year-1, year-3, year-5, and year-10 establishment survival rates, primary risk drivers, and source attribution. Data is derived from BLS Business Employment Dynamics (BED) longitudinal establishment-survival data, adjusted by sector-specific industry factors (capital intensity, regulation, customer concentration, demand stability).

All-sector baselines from BLS BED: ~80% survive year 1, ~62% survive year 3, ~50% survive year 5, ~35% survive year 10. The variance by sector is the story — utilities and finance sit at the top (year 5 ~80%), drinking places and book stores sit at the bottom (year 5 ~18-24%).

Sector summary (averaged across NAICS-4 codes within sector)

NAICS-2	Sector	#codes	Y1	Y3	Y5	Y10
11	Agriculture, Forestry, Fishing and Hunting	19	80.1%	63.1%	49.4%	32.8%
21	Mining, Quarrying, Oil and Gas Extraction	5	70.0%	51.4%	36.8%	22.2%
22	Utilities	3	92.7%	86.7%	80.7%	70.0%
23	Construction	10	78.2%	61.5%	48.6%	32.9%
31-33	Manufacturing	86	76.0%	58.7%	45.1%	29.5%
42	Wholesale Trade	19	77.6%	60.9%	48.3%	32.8%
44-45	Retail Trade	28	70.7%	51.8%	38.1%	23.5%
48-49	Transportation and Warehousing	29	76.2%	59.4%	46.8%	32.4%
51	Information	11	74.4%	57.1%	44.0%	29.3%
52	Finance and Insurance	10	83.0%	70.4%	60.2%	47.4%
53	Real Estate and Rental and Leasing	8	78.0%	61.8%	49.0%	34.2%
54	Professional, Scientific, Technical Services	9	77.6%	61.3%	48.7%	33.8%
56	Administrative + Support / Waste Management	11	77.3%	60.5%	47.8%	33.1%
61	Educational Services	7	81.1%	68.3%	57.7%	43.7%
62	Health Care and Social Assistance	18	81.7%	67.4%	55.8%	41.6%

NAICS-2	Sector	#codes	Y1	Y3	Y5	Y10
71	Arts, Entertainment, and Recreation	9	72.2%	54.0%	40.7%	26.0%
72	Accommodation and Food Services	6	71.0%	52.0%	38.2%	23.3%
81	Other Services (except Public Administration)	14	77.3%	61.6%	49.1%	34.4%

Most-distressed industries (lowest 5-year survival)

These are the categories where BizRadar reports must lead with realistic failure context and conservative cost-band recommendations.

NAICS-4	Title	Y1	Y3	Y5	Y10
7224	Drinking Places (Alcoholic Beverages)	56%	32%	18%	6%
4521	Department Stores	58%	36%	22%	8%
4543	Direct Selling Establishments (MLM)	60%	36%	22%	8%
4853	Taxi and Limousine Service	60%	36%	22%	8%
3346	Magnetic and Optical Media	60%	38%	24%	10%
4512	Book Stores and News Dealers	60%	38%	24%	10%
5111	Newspaper, Periodical, Book, Directory Publishers	60%	38%	24%	10%
7115	Independent Artists, Writers, Performers	60%	38%	24%	10%
2121	Coal Mining	60%	40%	25%	12%
4482	Shoe Stores	62%	40%	26%	12%
3152	Cut and Sew Apparel Manufacturing	64%	44%	28%	14%
4431	Electronics and Appliance Stores	64%	42%	28%	14%
4481	Clothing Stores	64%	42%	28%	14%
4511	Sporting Goods, Hobby, Musical Instrument Stores	64%	42%	28%	14%
4532	Office Supplies, Stationery, Gift Stores	64%	42%	28%	14%

Most-stable industries (highest 5-year survival)

These categories are slow-changing and have lower revenue-growth opportunity surface — BizRadar reports for these tend to lean on regulatory + acquisition + named-program playbook entries rather than fast-cycle marketing tactics.

NAICS-4	Title	Y1	Y3	Y5	Y10
4911	Postal Service	95%	90%	84%	75%

NAICS-4	Title	Y1	Y3	Y5	Y10
2213	Water, Sewage and Other Systems	94%	88%	82%	72%
2211	Electric Power Gen, Transmission, Distribution	92%	86%	80%	68%
2212	Natural Gas Distribution	92%	86%	80%	70%
5221	Depository Credit Intermediation	92%	86%	80%	70%
5232	Securities and Commodity Exchanges	92%	86%	80%	70%
6111	Elementary and Secondary Schools	92%	86%	80%	70%
6113	Colleges, Universities, Professional Schools	92%	86%	80%	68%
6221	General Medical and Surgical Hospitals	92%	86%	80%	70%
4862	Pipeline Transportation of Natural Gas	92%	84%	76%	64%
5241	Insurance Carriers	90%	82%	74%	62%
6112	Junior Colleges	90%	82%	74%	60%
6211	Offices of Physicians	90%	82%	74%	60%
4861	Pipeline Transportation of Crude Oil	90%	80%	72%	60%
4821	Rail Transportation	88%	78%	70%	58%

Full database by sector

Every NAICS-4 code with risk drivers + source attribution. Engineering teams use these values to calibrate cost-band recommendations + report tone for each sector.

Sector 11 — Agriculture, Forestry, Fishing and Hunting

Code	Title	Y1/Y3/Y5/Y10	Risk drivers
1111	Oilseed and Grain Farming	88/75/65/50	commodity prices, weather, capex
1112	Vegetable and Melon Farming	82/65/52/36	weather, labor, thin margins
1113	Fruit and Tree Nut Farming	86/72/60/44	multi-yr capex, climate, water rights
1114	Greenhouse, Nursery, Floriculture	80/60/45/28	labor, energy, demand cycles
1119	Other Crop Farming	84/70/58/40	commodity, subsidy policy, climate
1121	Cattle Ranching and Farming	86/72/60/42	feed cost, beef/dairy prices, land
1122	Hog and Pig Farming	82/65/52/35	feed prices, disease (PED, ASF), cycles
1123	Poultry and Egg Production	84/68/56/38	integrator dynamics, HPAI, feed
1124	Sheep and Goat Farming	78/60/45/28	niche, predator/climate, thin margins

Code	Title	Y1/Y3/Y5/Y1 0	Risk drivers
1125	Aquaculture	75/55/40/22	capex, water/permit, disease
1129	Other Animal Production	76/58/42/25	niche, capex, predator/disease
1131	Timber Tract Operations	88/76/65/48	multi-decade capex, prices, fire/climate
1132	Forest Nurseries + Forest Product Gathering	78/60/45/28	niche demand, seasonal, climate
1133	Logging	76/58/42/25	mill demand, safety/comp, equipment
1141	Fishing	70/50/35/20	quota/permit, fuel, safety/insurance
1142	Hunting and Trapping	68/48/32/18	niche, season, permit
1151	Support Activities for Crop Production	82/65/52/36	ag cycle, labor, consolidation
1152	Support Activities for Animal Production	80/62/48/32	vet shortage, ag cycle, rural labor
1153	Support Activities for Forestry	78/60/45/28	mill demand, equipment, safety

Sector 21 — Mining, Quarrying, Oil and Gas Extraction

Code	Title	Y1/Y3/Y5/Y1 0	Risk drivers
2111	Oil and Gas Extraction	70/50/35/20	commodity volatility, capex, depletion
2121	Coal Mining	60/40/25/12	secular decline, environmental, capex
2122	Metal Ore Mining	72/55/40/25	commodity, capex, exploration risk
2123	Nonmetallic Mineral Mining and Quarrying	80/64/52/36	construction demand, reserve, permit
2131	Support Activities for Mining	68/48/32/18	oil cycles, rig count, concentration

Sector 22 — Utilities

Code	Title	Y1/Y3/Y5/Y1 0	Risk drivers
2211	Electric Power Gen, Transmission, Distribution	92/86/80/68	regulated, fuel mix, capex
2212	Natural Gas Distribution	92/86/80/70	regulated, flow-through, capex
2213	Water, Sewage and Other Systems	94/88/82/72	regulated, capex, essential

Sector 23 — Construction

Code	Title	Y1/Y3/Y5/Y1 0	Risk drivers
2361	Residential Building Construction	76/58/45/30	housing cycle, sub/labor, cash flow
2362	Nonresidential Building Construction	78/62/50/35	CRE cycle, pipeline, bonding
2371	Utility System Construction	84/70/58/42	public pipeline, union/bonding, equipment
2372	Land Subdivision	72/52/38/22	development cycle, capital/entitlement, absorption
2373	Highway, Street, and Bridge Construction	82/68/56/40	DOT cycle, bonding, equipment
2379	Other Heavy and Civil Engineering Construction	80/65/52/36	public funding, cert, union
2381	Foundation, Structure, Building Exterior Contractors	76/58/45/30	ground-up cycle, labor, bonding
2382	Building Equipment Contractors (HVAC/elec/plumbing)	80/64/52/36	cyclical demand, tradesmen, parts cost
2383	Building Finishing Contractors	76/58/44/28	residential cycle, labor, materials
2389	Other Specialty Trade Contractors	78/60/46/30	trade-specific cycle, labor, equipment

Sector 31-33 — Manufacturing

Code	Title	Y1/Y3/Y5/Y1 0	Risk drivers
3111	Animal Food Manufacturing	84/70/58/42	feed prices, B2B concentration, regulation
3112	Grain and Oilseed Milling	86/72/60/44	commodity flow, B2B contracts, capex
3113	Sugar and Confectionery Manufacturing	82/68/55/40	retail/foodservice, sugar policy, private label
3114	Fruit and Vegetable Preserving and Specialty Food	80/65/52/36	retail consolidation, commodity, co-pack
3115	Dairy Product Manufacturing	84/70/58/42	milk price, co-op dynamics, capex
3116	Animal Slaughtering and Processing	78/60/46/30	USDA FSIS, capex, consolidation
3117	Seafood Product Preparation	76/58/44/28	supply volatility, cold chain, thin margins
3118	Bakeries and Tortilla Manufacturing	78/60/46/30	retail channel, commodity, consolidation
3119	Other Food Manufacturing (snack, coffee, condiment)	76/58/45/28	retail/slotting, private label, commodity
3121	Beverage Manufacturing	74/56/42/26	retail channel, distributor, capex (craft)
3122	Tobacco Manufacturing	80/65/52/38	secular decline, regulation, consolidation

Code	Title	Y1/Y3/Y5/Y1 0	Risk drivers
3131	Fiber, Yarn, Thread Mills	70/50/35/20	secular offshore, capex, demand
3132	Fabric Mills	68/48/32/18	secular decline, capex, foreign comp
3133	Textile and Fabric Finishing Mills	70/50/35/20	secular decline, environmental, foreign
3141	Textile Furnishings Mills	72/52/38/22	housing cycle, foreign, retail consolidation
3149	Other Textile Product Mills	70/50/35/20	niche, foreign, fashion cycle
3151	Apparel Knitting Mills	66/46/30/16	offshore shift, capex, fashion
3152	Cut and Sew Apparel Manufacturing	64/44/28/14	offshore shift, fashion, retail consolidation
3159	Apparel Accessories	66/46/30/16	offshore, fashion, thin margins
3161	Leather and Hide Tanning	68/48/32/18	offshore shift, environmental, demand
3162	Footwear Manufacturing	66/46/30/16	offshore, fashion, capex
3169	Other Leather and Allied Product	68/48/32/18	niche, offshore, fashion
3211	Sawmills and Wood Preservation	76/58/44/28	housing cycle, commodity, capex
3212	Veneer, Plywood, Engineered Wood	78/60/46/30	housing cycle, commodity, capex
3219	Other Wood Product Manufacturing	76/58/44/28	housing cycle, foreign, capex
3221	Pulp, Paper, and Paperboard Mills	84/70/58/42	digital shift, capex, environmental
3222	Converted Paper Product Manufacturing	78/62/50/34	e-comm packaging, commodity, consolidation
3231	Printing and Related Support	70/52/38/22	digital shift, consolidation, demand erosion
3241	Petroleum and Coal Products	82/68/55/40	oil cycles, capex, energy transition
3251	Basic Chemical Manufacturing	80/65/52/36	commodity cycle, capex, environmental
3252	Resin, Synthetic Rubber, Fiber	78/62/50/34	commodity, capex, foreign
3253	Pesticide, Fertilizer, Agricultural Chemical	80/64/52/36	ag commodity, environmental, capex
3254	Pharmaceutical and Medicine Manufacturing	86/74/62/48	FDA pathway, capex, patents
3255	Paint, Coating, and Adhesive	78/62/50/34	commodity, B2B/retail, environmental
3256	Soap, Cleaning Compound	76/58/46/30	retail/private label, consumer trends, commodity
3259	Other Chemical Product	78/62/50/34	niche, commodity, environmental
3261	Plastics Product Manufacturing	76/58/44/28	commodity, foreign, B2B concentration

Code	Title	Y1/Y3/Y5/Y1 0	Risk drivers
3262	Rubber Product Manufacturing	76/58/44/28	commodity, capex, foreign
3271	Clay Product and Refractory	78/60/46/30	construction cycle, capex, foreign
3272	Glass and Glass Product	76/58/44/28	construction/auto cycles, capex, foreign
3273	Cement and Concrete Product	84/70/58/42	construction cycle, capex, local moat
3274	Lime and Gypsum Product	84/70/58/42	construction, capex/transport, commodity
3279	Other Nonmetallic Mineral Product	76/58/44/28	niche, construction, capex
3311	Iron and Steel Mills and Ferroalloy	78/60/46/30	commodity/tariff, capex, foreign
3312	Steel Product from Purchased Steel	76/58/44/28	flow-through, B2B concentration, foreign
3313	Alumina and Aluminum Production	78/60/46/30	commodity/energy, capex, foreign
3314	Nonferrous Metal Production	76/58/44/28	commodity, capex, foreign
3315	Foundries	74/56/42/26	B2B concentration, capex/skilled labor, foreign
3321	Forging and Stamping	76/58/44/28	B2B concentration, auto/aero cycles, capex
3322	Cutlery and Handtool Manufacturing	72/54/40/24	retail channel, foreign, commodity
3323	Architectural and Structural Metals	78/60/46/30	construction cycle, commodity, project-based
3324	Boiler, Tank, and Shipping Container	80/64/52/36	B2B project-based, commodity, capex
3325	Hardware Manufacturing	74/56/42/26	retail/B2B, foreign, commodity
3326	Spring and Wire Product	76/58/44/28	B2B concentration, commodity, foreign
3327	Machine Shops; Turned Product; Screw, Nut, Bolt	76/58/45/28	B2B concentration, capex/skilled labor, pricing
3328	Coating, Engraving, Heat Treating	78/60/46/30	B2B, capex/cert (NADCAP, AS9100), environmental
3329	Other Fabricated Metal Product	76/58/44/28	niche B2B, commodity, foreign
3331	Ag, Construction, Mining Machinery	78/62/50/34	ag/construction/mining cycles, B2B, capex
3332	Industrial Machinery Manufacturing	76/58/44/28	B2B concentration, foreign, capex
3333	Commercial and Service Industry Machinery	76/58/44/28	B2B/retail, foreign, capex
3334	Ventilation, Heating, A/C, Refrigeration	80/64/52/36	construction/replacement cycles, commodity, consolidation

Code	Title	Y1/Y3/Y5/Y1 0	Risk drivers
3335	Metalworking Machinery Manufacturing	76/58/44/28	B2B concentration, capex, foreign
3336	Engine, Turbine, Power Transmission	78/62/50/34	B2B concentration, capex, energy transition
3339	Other General Purpose Machinery	76/58/44/28	B2B niche, capex, foreign
3341	Computer and Peripheral Equipment	70/50/36/20	Asian shift, tech turnover, commodity
3342	Communications Equipment Manufacturing	72/54/40/24	B2B/telecom cycles, tech turnover, capex
3343	Audio and Video Equipment Manufacturing	68/48/32/18	Asian shift, consumer trends, turnover
3344	Semiconductor and Electronic Component	76/60/46/30	cyclical, capex, foreign
3345	Navigational, Measuring, Electromedical, Control Instruments	82/68/56/42	regulated demand (medical, defense), B2B, capex/cert
3346	Magnetic and Optical Media	60/38/24/10	secular decline (digital), turnover, demand erosion
3351	Electric Lighting Equipment	74/56/42/26	LED transition, commodity, foreign
3352	Household Appliance Manufacturing	76/58/44/28	retail channel, foreign, consolidation
3353	Electrical Equipment Manufacturing	78/62/50/34	B2B/utility cycles, capex, commodity
3359	Other Electrical Equipment	76/58/44/28	B2B niche, commodity, foreign
3361	Motor Vehicle Manufacturing	80/65/52/38	cyclical demand, capex, tier-1 dynamics
3362	Motor Vehicle Body and Trailer	76/58/44/28	cyclical demand, B2B/retail, capex
3363	Motor Vehicle Parts Manufacturing	78/62/50/34	OEM concentration, tier-2 dynamics, EV transition
3364	Aerospace Product and Parts	84/72/60/46	program cycles 10-30 yr, AS9100/NADCAP, capex
3365	Railroad Rolling Stock Manufacturing	80/64/52/36	B2B/freight cycle, capex, consolidation
3366	Ship and Boat Building	72/54/40/24	cyclical, capex, specialty cert
3369	Other Transportation Equipment	74/56/42/26	niche, capex, cyclicity
3371	Household, Institutional Furniture, Kitchen Cabinet	70/52/38/22	housing cycle, foreign, retail
3372	Office Furniture (incl. Fixtures)	74/56/42/26	commercial cycle, WFH, B2B
3379	Other Furniture Related Product	70/52/38/22	niche, foreign, cyclical

Code	Title	Y1/Y3/Y5/Y1 0	Risk drivers
3391	Medical Equipment and Supplies	84/72/60/46	FDA/ISO 13485, capex/cert, reimbursement
3399	Other Miscellaneous Manufacturing	74/56/42/26	niche, commodity, foreign

Sector 42 — Wholesale Trade

Code	Title	Y1/Y3/Y5/Y1 0	Risk drivers
4231	Motor Vehicle and Parts Wholesalers	78/62/50/36	auto OEM cycles, inventory/capex, consolidation
4232	Furniture and Home Furnishing Wholesalers	74/56/42/28	housing cycle, foreign, retail
4233	Lumber and Construction Materials Wholesalers	78/62/50/34	construction cycle, commodity, big-box (HD/Lowes)
4234	Professional and Commercial Equipment Wholesalers	78/62/50/34	B2B cycle, tech evolution, consolidation
4235	Metal and Mineral Wholesalers	78/62/50/34	commodity volatility, B2B concentration, consolidation
4236	Household Appliances and Electrical Wholesalers	76/58/44/28	construction/replacement, big-box, foreign
4237	Hardware, Plumbing, Heating Wholesalers	80/64/52/36	construction/trade cycle, big-box, consolidation
4238	Machinery, Equipment, Supplies Wholesalers	78/62/50/34	B2B cycle, commodity, consolidation
4239	Miscellaneous Durable Goods Wholesalers	76/58/44/28	niche, foreign, consolidation
4241	Paper and Paper Product Wholesalers	76/58/44/28	digital shift, consolidation, demand
4242	Drugs and Druggists' Sundries Wholesalers	86/74/64/50	regulated (DEA), 3-major consolidation, capex
4243	Apparel, Piece Goods Wholesalers	70/50/36/22	fashion cycle, foreign, retail consolidation
4244	Grocery and Related Product Wholesalers	80/64/52/36	retail consolidation, foodservice cycles, pressure
4245	Farm Product Raw Material Wholesalers	78/62/50/34	commodity volatility, ag cycles, consolidation

Code	Title	Y1/Y3/Y5/Y1 0	Risk drivers
4246	Chemical and Allied Products Wholesalers	80/64/52/36	B2B concentration, environmental/regulation, consolidation
4247	Petroleum Products Wholesalers	78/62/50/34	commodity volatility, consolidation, energy transition
4248	Beer, Wine, Distilled Beverage Wholesalers	84/70/58/42	3-tier system, state-protected, consolidation
4249	Miscellaneous Nondurable Goods Wholesalers	76/58/44/28	niche, foreign, consolidation
4251	Wholesale Electronic Markets and Brokers	70/50/36/22	platform shifts, consolidation, commodity

Sector 44-45 — Retail Trade

Code	Title	Y1/Y3/Y5/Y1 0	Risk drivers
4411	Automobile Dealers	78/62/50/36	OEM agreements, cyclical, EV transition
4412	Other Motor Vehicle Dealers	72/54/40/24	niche, cyclical, consolidation
4413	Auto Parts, Accessories, Tire Stores	76/58/44/28	big-box/online, EV transition, consolidation
4421	Furniture Stores	70/50/36/22	housing cycle, online (Wayfair), consolidation
4422	Home Furnishings Stores	68/48/32/18	housing cycle, online, consolidation
4431	Electronics and Appliance Stores	64/42/28/14	big-box/online (BBY, Amazon), tech turnover, collapse
4441	Building Material and Supplies Dealers	78/62/50/36	big-box (HD/Lowes), construction cycle, consolidation
4442	Lawn and Garden Equipment and Supplies	70/52/38/24	big-box, seasonal, online
4451	Grocery Stores	78/62/50/36	big-box/online (Walmart, Kroger, Amazon), thin margins
4452	Specialty Food Stores	70/52/38/22	big-box encroachment, thin margins, consolidation
4453	Beer, Wine, and Liquor Stores	80/64/52/38	state-regulated, license-protected, big-box (Costco)
4461	Health and Personal Care Stores	76/58/44/28	CVS/Walgreens/RiteAid, online (Amazon), consolidation

Code	Title	Y1/Y3/Y5/Y1 0	Risk drivers
4471	Gasoline Stations	80/64/52/38	fuel margins, EV transition, c-store transition
4481	Clothing Stores	64/42/28/14	fast-fashion/online (SHEIN, Amazon), mall decline
4482	Shoe Stores	62/40/26/12	online/DTC/Amazon, mall decline, consolidation
4483	Jewelry, Luggage, Leather Goods Stores	66/46/30/16	online/DTC/Amazon, mall decline, consolidation
4511	Sporting Goods, Hobby, Musical Instrument Stores	64/42/28/14	online/Amazon, specialty erosion, consolidation
4512	Book Stores and News Dealers	60/38/24/10	Amazon/digital shift, mall decline, collapse
4521	Department Stores	58/36/22/8	Amazon/DTC, mall decline, category collapse
4522	Other General Merchandise (Walmart, Target)	84/72/60/46	big-box/omnichannel, consolidation, dominance
4529	Other General Merchandise (warehouse, dollar)	78/62/50/36	warehouse club/dollar competition, consolidation
4531	Florists	68/48/32/18	online (1-800-Flowers, FTD), thin margins
4532	Office Supplies, Stationery, Gift Stores	64/42/28/14	Amazon/Office Depot/Staples, erosion, consolidation
4533	Used Merchandise Stores (thrift, consignment)	76/58/44/28	secular thrift growth, platform comp (Poshmark), inventory
4539	Other Miscellaneous Store Retailers	70/52/38/22	niche, online, consolidation
4541	Electronic Shopping and Mail-Order	70/50/36/20	Amazon dominance, algorithm dynamics, capex/DTC
4542	Vending Machine Operators	76/58/44/28	contract/access, consolidation, cashless
4543	Direct Selling Establishments (MLM)	60/36/22/8	FTC pressure, high turnover, channel dynamics

Sector 48-49 — Transportation and Warehousing

Code	Title	Y1/Y3/Y5/Y1 0	Risk drivers
4811	Scheduled Air Transportation	70/50/36/20	fuel/capex, cyclical, consolidation
4812	Nonscheduled Air Transportation	64/42/28/14	fuel/capex, cyclical, consolidation

Code	Title	Y1/Y3/Y5/Y1 0	Risk drivers
4821	Rail Transportation	88/78/70/58	regulated/protected, capex, Class I
4831	Deep Sea, Coastal, Great Lakes Water	76/58/44/28	fuel/capex, cyclical, consolidation
4832	Inland Water Transportation	80/64/52/36	barge/commodity flow, capex, consolidation
4841	General Freight Trucking	74/56/42/26	driver shortage, fuel volatility, spot freight
4842	Specialized Freight Trucking	76/58/44/28	specialty trailer cycles, driver retention, fuel
4851	Urban Transit Systems	86/74/64/50	public-funded/regulated, capex/ridership, subsidy
4852	Interurban and Rural Bus	76/58/44/28	ridership decline, fuel/capex, consolidation
4853	Taxi and Limousine Service	60/36/22/8	Uber/Lyft disruption, retention, medallion collapse
4854	School and Employee Bus Transportation	84/70/58/42	contract-driven, driver retention, consolidation
4855	Charter Bus Industry	64/42/28/14	cyclical demand, fuel/capex, consolidation
4859	Other Transit and Ground Passenger	64/42/28/14	ridesharing disruption, niche, consolidation
4861	Pipeline Transportation of Crude Oil	90/80/72/60	regulated, long-term contract, capex
4862	Pipeline Transportation of Natural Gas	92/84/76/64	regulated/protected, long-term contract, capex
4869	Other Pipeline Transportation	88/76/68/56	regulated/long-term, capex, commodity
4871	Scenic and Sightseeing Transportation, Land	70/50/36/20	tourism cycles, seasonal, capex
4872	Scenic and Sightseeing Transportation, Water	68/48/32/18	tourism cycles, seasonal, capex/insurance
4879	Scenic and Sightseeing, Other	66/46/30/16	niche tourism, seasonal, capex
4881	Support Activities for Air Transportation	76/58/44/28	airline cycles, consolidation, fuel/cost
4882	Support Activities for Rail	82/68/56/40	Class I dynamics, capex, consolidation
4883	Support Activities for Water	76/58/44/28	port/commodity flow, capex, consolidation
4884	Support Activities for Road	74/56/42/26	trucking cycle, consolidation, tech disruption

Code	Title	Y1/Y3/Y5/Y1 0	Risk drivers
4885	Freight Transportation Arrangement (brokers)	70/50/36/22	margin compression, digital broker disruption, consolidation
4889	Other Support Activities for Transportation	74/56/42/26	niche, consolidation, cycles
4911	Postal Service	95/90/84/75	gov entity (USPS), regulated, sustainability
4921	Couriers and Express Delivery Services	78/62/50/36	UPS/FedEx/Amazon dominance, consolidation, last-mile
4922	Local Messengers and Local Delivery	64/42/28/14	gig economy disruption, thin margins, consolidation
4931	Warehousing and Storage	84/70/58/44	e-commerce growth, capex, consolidation

Sector 51 — Information

Code	Title	Y1/Y3/Y5/Y1 0	Risk drivers
5111	Newspaper, Periodical, Book, Directory Publishers	60/38/24/10	digital shift, ad collapse, transformation
5112	Software Publishers (incl. SaaS)	72/54/40/26	startup churn, capex/GTM, category dynamics
5121	Motion Picture and Video Industries	72/54/40/24	project-based, consolidation, streaming
5122	Sound Recording Industries	70/52/38/22	streaming transition, 3-major consolidation, margins
5151	Radio and Television Broadcasting	80/64/52/38	secular ad shift, consolidation, streaming pressure
5152	Cable and Other Subscription Programming	78/62/50/36	cord-cutting, streaming pressure, consolidation
5173	Wired/Wireless Telecommunications Carriers	84/70/58/44	regulated/capex, oligopoly, tech transition
5174	Satellite Telecommunications	80/64/52/38	capex, tech transition (Starlink), consolidation
5179	Other Telecommunications	72/54/40/24	niche, capex, consolidation
5182	Computing Infrastructure, Data Processing, Hosting	78/62/50/36	cloud (AWS/Azure/GCP), consolidation, capex
5191	Web Search Portals, Libraries, Archives	72/54/40/24	Google dominance, category dynamics, business model

Sector 52 — Finance and Insurance

Code	Title	Y1/Y3/Y5/Y10	Risk drivers
5221	Depository Credit Intermediation	92/86/80/70	regulated/insured, capital req, consolidation
5222	Nondepository Credit Intermediation	78/62/50/36	regulatory cycles, rate volatility, consolidation
5223	Activities Related to Credit Intermediation	70/50/36/22	mortgage cycles, rate volatility, consolidation
5231	Securities Brokerage	78/62/50/36	market cycles, consolidation, fee compression
5232	Securities and Commodity Exchanges	92/86/80/70	regulated/protected, oligopoly, capex
5239	Other Financial Investment Activities	80/64/52/38	investor cycles, fee dynamics, consolidation
5241	Insurance Carriers	90/82/74/62	regulated/reserves, capital, consolidation
5242	Agencies, Brokerages, Insurance Activities	84/72/60/46	carrier relationships, consolidation, fees
5251	Insurance and Employee Benefit Funds	88/78/70/58	regulated/protected, capital/reserve, consolidation
5259	Other Investment Pools and Funds	78/62/50/36	investor cycles, fee compression, consolidation

Sector 53 — Real Estate and Rental and Leasing

Code	Title	Y1/Y3/Y5/Y10	Risk drivers
5311	Lessors of Real Estate	86/74/62/48	rent cycles, capital/leverage, consolidation
5312	Offices of Real Estate Agents and Brokers	78/62/50/36	agent productivity, cyclical, consolidation
5313	Activities Related to Real Estate (property mgmt)	80/64/52/38	contract concentration, cycle, consolidation
5321	Automotive Equipment Rental and Leasing	76/58/44/28	fleet capex, cyclical, consolidation
5322	Consumer Goods Rental	70/52/38/22	niche, capex, consolidation
5323	General Rental Centers (incl. self-storage)	80/64/52/38	real estate cycles, REIT consolidation, demand

Code	Title	Y1/Y3/Y5/Y1 0	Risk drivers
5324	Commercial and Industrial Machinery Rental	76/58/44/28	B2B/capex cycles, consolidation, capex
5331	Lessors of Nonfinancial Intangible Assets	78/62/50/36	IP licensing, consolidation, patent cycles

Sector 54 — Professional, Scientific, Technical Services

Code	Title	Y1/Y3/Y5/Y1 0	Risk drivers
5411	Legal Services	80/64/52/38	attorney retention, practice cycles, consolidation
5412	Accounting, Tax Prep, Bookkeeping, Payroll	84/72/60/46	recurring revenue, consolidation, tech transition
5413	Architectural, Engineering, Related	80/64/52/38	project-based, BD discipline, consolidation
5414	Specialized Design Services	70/52/38/22	project-based, consolidation, commoditization
5415	Computer Systems Design and Related	76/58/44/28	client concentration, tech transition, consolidation
5416	Management, Scientific, Technical Consulting	78/62/50/36	relationship-driven, project-based, consolidation
5417	Scientific R&D; Services	80/64/52/36	contract/grant funding, capital/talent, consolidation
5418	Advertising, PR, Related	70/52/38/22	client concentration, channel transformation, consolidation
5419	Other Professional Services (incl. veterinary)	80/64/52/38	niche specialty, consolidation, cycle dynamics

Sector 56 — Administrative + Support / Waste Management

Code	Title	Y1/Y3/Y5/Y1 0	Risk drivers
5611	Office Administrative Services	76/58/44/28	contract concentration, consolidation, tech transition
5612	Facilities Support Services	80/64/52/38	contract concentration, labor, consolidation
5613	Employment Services (staffing)	76/58/44/28	cyclical, MSP/VMS pressure, consolidation

Code	Title	Y1/Y3/Y5/Y1 0	Risk drivers
5614	Business Support Services	74/56/42/26	contract concentration, consolidation, tech displacement
5615	Travel Arrangement and Reservation	64/42/28/14	platform disruption (Booking, Expedia), consolidation
5616	Investigation and Security Services	78/62/50/36	contract concentration, consolidation, labor cost
5617	Services to Buildings and Dwellings	78/62/50/36	contract concentration, labor, consolidation
5619	Other Support Services	74/56/42/26	niche, consolidation, cycle dynamics
5621	Waste Collection	84/70/58/44	contract/route density, consolidation (WM, Republic), capex
5622	Waste Treatment and Disposal	86/74/64/50	regulated/capex, consolidation, permit/entitlement
5629	Remediation and Other Waste Management	80/64/52/38	project-based, regulation/cert, consolidation

Sector 61 — Educational Services

Code	Title	Y1/Y3/Y5/Y1 0	Risk drivers
6111	Elementary and Secondary Schools	92/86/80/70	regulated/accredited, enrollment cycles, community
6112	Junior Colleges	90/82/74/60	public funding, regulated, enrollment cycles
6113	Colleges, Universities, Professional Schools	92/86/80/68	accreditation/endowment, enrollment cycles, public/private mix
6114	Business Schools and Computer/Management Training	70/52/38/22	cyclical demand, tech transition, bootcamp consolidation
6115	Technical and Trade Schools	76/58/44/28	Title IV compliance, accreditation, regulatory pressure
6116	Other Schools and Instruction	70/52/38/22	niche demand, consolidation, fragmented competition
6117	Educational Support Services	78/62/50/36	contract/cycle, tech transition, consolidation

Sector 62 — Health Care and Social Assistance

Code	Title	Y1/Y3/Y5/Y10	Risk drivers
6211	Offices of Physicians	90/82/74/60	credentialing/insurance contracts, consolidation, reimbursement
6212	Offices of Dentists	88/78/70/56	credentialing/DSO consolidation, capex, reimbursement
6213	Offices of Other Health Practitioners	80/64/52/38	niche specialty, reimbursement variability, consolidation
6214	Outpatient Care Centers	86/74/62/48	regulated/accredited, reimbursement, consolidation
6215	Medical and Diagnostic Laboratories	86/74/62/48	regulated/cert (CLIA), reimbursement, consolidation
6216	Home Health Care Services	76/58/44/28	Medicare reimbursement, labor/retention, consolidation
6219	Other Ambulatory Health Care Services	80/64/52/38	niche specialty, reimbursement, consolidation
6221	General Medical and Surgical Hospitals	92/86/80/70	accreditation/capex, community-essential, consolidation
6222	Psychiatric and Substance Abuse Hospitals	84/70/58/44	accreditation/reimbursement, capex, regulated
6223	Specialty Hospitals	86/74/62/48	specialty accreditation, reimbursement, capex
6231	Nursing Care Facilities (SNFs)	80/64/52/38	Medicare/Medicaid reimbursement, staffing crisis, regulation
6232	Residential IDD/Mental Health Facilities	78/62/50/36	public-funded, staffing, regulation
6233	CCRCs and Assisted Living	82/68/56/42	demographic tailwind, staffing, capex/reimbursement
6239	Other Residential Care Facilities	76/58/44/28	public-funded, staffing, regulation
6241	Individual and Family Services	76/58/44/28	public-funded/nonprofit, staffing, cycle dynamics
6242	Community Food and Housing, Emergency Relief	80/64/52/38	nonprofit/grant-funded, community-essential, cycle dynamics
6243	Vocational Rehabilitation Services	80/64/52/38	public-funded, regulated, cycle dynamics
6244	Child Day Care Services	70/52/38/22	staffing crisis, regulation/cert, thin margins

Sector 71 — Arts, Entertainment, and Recreation

Code	Title	Y1/Y3/Y5/Y1 0	Risk drivers
7111	Performing Arts Companies	64/42/28/14	donor/ticket cycles, thin margins, consolidation
7112	Spectator Sports	80/64/52/38	league protection, capex, consolidation
7113	Promoters of Performing Arts, Sports, Events	64/42/28/14	cycle volatility, event risk, consolidation
7114	Agents and Managers for Talent	70/52/38/22	talent dynamics, cycle, consolidation
7115	Independent Artists, Writers, Performers	60/38/24/10	gig economy, sole-prop failure, cycle volatility
7121	Museums, Historical Sites, Similar	86/74/62/48	nonprofit/endowment, community-supported, stability
7131	Amusement Parks and Arcades	76/58/44/28	capex/cycles, consolidation, demographics
7132	Gambling Industries	80/64/52/38	regulated/licensed, capex, consolidation
7139	Other Amusement and Recreation	70/52/38/22	cycle volatility, capex, consolidation

Sector 72 — Accommodation and Food Services

Code	Title	Y1/Y3/Y5/Y1 0	Risk drivers
7211	Traveler Accommodation (hotels)	76/58/44/28	cyclical demand, capex, STR competition
7212	RV Parks and Recreational Camps	78/62/50/36	seasonal, capex, consolidation
7213	Rooming and Boarding Houses, Dormitories	76/58/44/28	niche markets, cycle, consolidation
7223	Special Food Services (caterers)	70/52/38/22	cycle volatility, event risk, consolidation
7224	Drinking Places (Alcoholic Beverages)	56/32/18/6	highest failure category, thin margins, lifestyle shifts
7225	Restaurants and Other Eating Places	70/50/35/20	thin margins, labor, location/lease

Sector 81 — Other Services (except Public Administration)

Code	Title	Y1/Y3/Y5/Y1 0	Risk drivers
8111	Automotive Repair and Maintenance	78/62/50/36	EV transition, parts costs, consolidation
8112	Electronic and Precision Equipment Repair	70/52/38/22	tech turnover, replace-don't-fix, consolidation

Code	Title	Y1/Y3/Y5/Y1 0	Risk drivers
8113	Commercial and Industrial Machinery Repair	78/62/50/36	B2B contract, technician shortage, consolidation
8114	Personal and Household Goods Repair	66/46/30/16	replace-don't-fix, online, consolidation
8121	Personal Care Services (salons, barbers, nails, spa)	72/54/40/24	gig/booth-rental dynamics, lease cost, consolidation
8122	Death Care Services (funeral homes, cemeteries)	86/74/62/48	regulated/capex, community/family, consolidation (SCI)
8123	Drycleaning and Laundry Services	70/52/38/22	commercial transition, environmental, consolidation
8129	Other Personal Services (pet, photo, etc.)	70/52/38/22	niche markets, online, consolidation
8131	Religious Organizations	88/78/70/56	tax-exempt, community-supported, attendance dynamics
8132	Grantmaking and Giving Services	88/78/70/56	endowment/donor-funded, regulated, stable
8133	Social Advocacy Organizations	76/58/44/28	donor cycle, mission-driven, consolidation
8134	Civic and Social Organizations	78/62/50/36	membership dynamics, demographic shifts, consolidation
8139	Business, Professional, Labor Organizations	82/68/56/42	membership-supported, regulated, cycle stability
8141	Private Households	80/64/52/38	household employment dynamics, demographic, regulation

PART 4 — IMPLEMENTATION ROADMAP

Five phases, gated quality

BizRadar v4 ships across five sequential phases, each gated on quality + production metrics rather than calendar dates. The current production system (v3) handles hospitality (~33% of 1,464 business types) well; v4 adds the remaining 67% across all 18 in-scope NAICS sectors. Phases 1-2 protect existing v3 quality while building the universal scaffolding underneath. Phases 3-5 progressively add sector profiles in order of (a) market size, (b) ease of implementation, and (c) distance from v3 substrate.

Phase 0 — Master Router scaffolding

Duration estimate: 4-6 weeks

Goal

Add Layer 0.5 Master Router that classifies inputs by NAICS-2 + business model + sector_profile_id, returning the same hospitality logic for sector 72 inputs while preparing the substitution architecture for all other sectors.

Deliverables

- naicsRouter.js — exact-match table from NAICS-6 → sector_profile_id, with NAICS-4 fallback and NAICS-2 fallback
- businessTypeKeywordRouter.js — keyword-match for free-text business type queries (e.g. "coffee shop" → hospitality.cafe_quick_service)
- modelClassifier.js — classifies any business input into business_model: B2C_walk_in / B2C_appointment / B2C_subscription / B2B_relationship / B2B_transactional / B2B_subscription / mixed
- sectorProfile.js — registry mapping profile_id to its data sources, competitor radius, review categories, opportunity playbook, etc.
- outOfScopeHandler.js — early termination for sector 55 + 92 inputs with structured user message and refund/no-charge handling

Gates to advance from Phase 0

1. Master Router classifies 100% of v3 production inputs (hospitality) with zero behavior change vs current v3 — verified by replaying 1,000 production inputs
2. Master Router correctly classifies a sample of 200 inputs from each non-hospitality sector with > 92% accuracy on profile_id (manual evaluation)
3. Out-of-scope handler triggers correctly for sector 55 + 92 inputs
4. Performance: Master Router adds < 50ms latency to existing v3 path

Phase 1 — Hospitality + closest cousins (sectors 72, 44-45, 81)

Duration estimate: 8-12 weeks

Goal

Migrate v3 hospitality logic into hospitality.* sector profiles inside the new architecture, then add the closest-cousin profiles (Retail, Other Services personal care + repair) that share most of v3 substrate.

Deliverables

- hospitality.cafe_quick_service, hospitality.full_service_restaurant, hospitality.bar_nightlife, hospitality.lodging — all migrated from v3 with zero behavior change
- retail.specialty_brick_mortar, retail.ecommerce_dtc, retail.multi_channel — new profiles built on hospitality substrate
- other_services.personal_care, other_services.auto_repair — new profiles built on hospitality substrate
- Per-profile fixture set + golden output (ground-truth report) for regression testing

Gates to advance from Phase 1

1. All Phase 1 profiles pass quality gates: 12+ playbook entries, 6+ data sources, 5 cost scale bands, complete persona schema, complete cycle, complete issue library
2. Production traffic on Phase 1 profiles: > 95% of generated reports pass internal quality review (sample 50 / week)
3. No regression on v3 hospitality customer NPS (must remain $\geq$ baseline)
4. New-sector revenue attributable to retail.* + other_services.* > \$5k/mo by end of phase

Phase 2 — Recreation, Healthcare, Construction (sectors 71, 62, 23)

Duration estimate: 12-16 weeks

Goal

Add the medium-distance sectors that exercise B2C_appointment + B2B_relationship patterns at meaningful scale. These are bigger market sectors but require new substrate (healthcare reimbursement logic, construction trade-cert logic, recreation seasonality).

Deliverables

- recreation.fitness_studio, recreation.amusement_attraction, recreation.spectator_sports — recreation profiles
- healthcare.medical_practice, healthcare.dental_practice, healthcare.allied_health, healthcare.behavioral_health, healthcare.specialty_clinic — healthcare profiles
- construction.residential_trades, construction.commercial_gc, construction.civil_heavy — construction profiles
- Healthcare reimbursement + cert tracker (named insurers + Medicare + state Medicaid + named accreditors)
- Construction trade-cert tracker (named state license boards + named manufacturer-certs like NATE, GAF Master Elite, Carrier Factory Authorized)

Gates to advance from Phase 2

1. All Phase 2 profiles pass quality gates
2. Healthcare profiles tested against named reimbursement scenarios (Medicare-driven, commercial-insurance-driven, cash-pay) without producing factually wrong cost bands
3. Construction profiles tested against named trade-cert scenarios without hallucinating non-existent certs

4. New-sector revenue attributable to recreation.* + healthcare.* + construction.* > \$20k/mo by end of phase
 5. Customer-reported quality complaints in Phase 2 sectors < 4% of generated reports
-

Phase 3 — Real Estate, Finance, Professional (sectors 53, 52, 54)

Duration estimate: 12-16 weeks

Goal

Add the high-LTV professional + financial services sectors. These exercise B2B_relationship + regulated patterns that no Phase 0-2 profile touches. Highest premium \$/report potential and most M&A-active; sub-clusters in the spec.

Deliverables

- real_estate.brokerage, real_estate.property_management, real_estate.self_storage — real estate profiles
- finance.community_bank, finance.insurance_agency, finance.ria_wealth_management — finance profiles
- professional.legal, professional.accounting, professional.architecture_engineering, professional.consulting, professional.veterinary — professional profiles
- Regulated B2B substrate (named regulators FDIC + NCUA + FINRA + SEC + state bar + AICPA + NCARB + named licensing boards)
- B2B-specific substrate (named target accounts + RFP + COI + named-property-mgmt customer pipeline)

Gates to advance from Phase 3

1. All Phase 3 profiles pass quality gates
 2. Veterinary profile (54.E) reuses 70% of healthcare.medical_practice substrate without modification — verified by code-share metric
 3. Professional services profiles correctly handle bar advertising rules + accounting compliance + designation requirements without generating non-compliant marketing recommendations
 4. New-sector revenue attributable to real_estate.* + finance.* + professional.* > \$40k/mo by end of phase
-

Phase 4 — B2B-heavy (Manufacturing, Wholesale, Transportation, Admin)

Duration estimate: 16-20 weeks

Goal

Add the B2B-heavy operational backbone sectors. Manufacturing alone has 346 NAICS codes — the biggest by far. Each sub-cluster exercises long-cycle B2B-relationship + capability-based marketing + named-OEM-customer patterns.

Deliverables

- manufacturing.food_beverage_dtc, manufacturing.industrial_b2b, manufacturing.advanced_regulated — manufacturing profiles

- wholesale.durable_goods, wholesale.nondurable_food_grocery — wholesale profiles
- transportation.trucking_freight, transportation.courier_last_mile, transportation.warehousing_3pl — transportation profiles
- admin.staffing_agency, admin.facility_services, admin.security_pest, admin.waste_management — admin/waste profiles
- Named-cert tracker (ISO 9001, ISO 13485, AS9100, IATF 16949, NADCAP, CMMC, SOC 2, BSCAI CIMS, NPMA, etc.)
- Named-OEM relationship tracker (Boeing, Lockheed, GM, Ford, Stellantis, Toyota, Honda, named tier-1s)
- Federal program tracker (Buy America, BABA, BEAD, USDA ReConnect, GSA Schedule, DLA, MEP Center, CMMC C3PAO list)

Gates to advance from Phase 4

1. All Phase 4 profiles pass quality gates
2. Manufacturing.industrial_b2b correctly distinguishes between capability-based and commodity-pricing B2B work — verified by sample test
3. Federal program tracker maintains > 95% accuracy on grant program eligibility criteria
4. No false-named-cert hallucinations in 100 sampled Phase 4 reports
5. New-sector revenue attributable to manufacturing.* + wholesale.* + transportation.* + admin.* > \$80k/mo by end of phase

Phase 5 — Specialty long-tail (Information, Education, Mining/Utilities/Agriculture)

Duration estimate: 20-26 weeks

Goal

Add the specialty long-tail sectors that exercise patterns no other sector uses: SaaS valuation multiples (51.A), broadcast ad-rep dynamics (51.D), accreditation cycles (61.B), academic calendar cycles (61.A-D), commodity cycles (11, 21), regulated utilities (22).

Deliverables

- information.saas_software, information.media_publishing, information.telecom_isp, information.broadcasting_production — information profiles
- education.tutoring_test_prep, education.k12_private, education.bootcamps_online, education.trade_technical — education profiles
- agriculture.crop_farming, agriculture.livestock, agriculture.support_services — agriculture profiles
- mining.oil_gas, mining.aggregates, mining.support_services — mining profiles
- utilities.electric, utilities.gas, utilities.water — utilities profiles
- SaaS metric tracker (NRR, GRR, CAC, LTV, magic number, Rule of 40)
- Accreditation tracker (named accreditors NAIS + NEASC + ACCSC + ACCET + COE + AMS + AWSNA + NAEYC)
- Academic calendar tracker (admissions cycles, FAFSA cycle, AP/SAT/ACT cycles, accreditation visit cycles)

Final gates for v4 GA

1. All 18 in-scope sectors live with passing quality gates
 2. Master Router accuracy > 96% across all sectors (verified by 2,000-sample manual evaluation)
 3. Customer NPS by sector all $\geq$ baseline hospitality NPS
 4. No false-named-entity hallucinations in 200 sampled reports
 5. New-sector revenue across all non-hospitality sectors > \$250k/mo by end of phase
 6. Production system handles 4× current v3 volume without degradation
-

Cross-phase principles

1. Quality gates beat calendar dates. If a phase doesn't pass its gate, the next phase doesn't start. Production rollback procedures must be established before each phase ships.
2. Each sector profile must reuse substrate where possible. Veterinary should reuse 70% of healthcare. Wholesale should reuse 50% of retail + manufacturing. Reuse is verified by code-share metric, not just architectural intent.
3. Named entities must be verifiable at deployment time. Each sector profile ships with a verification script that confirms every named entity (cert body, software platform, federal program, association) is currently operating + has a working URL.
4. No phase ships if it would degrade v3 hospitality quality. Customer-facing v3 NPS is the floor — every phase release must hold or improve it.
5. Phases 4-5 are the ROI tail of the spec. Most reports will continue to come from Phase 0-3 sectors (hospitality + retail + healthcare + professional). Phases 4-5 are unlock + premium-positioning phases.

PART 5 — APPENDICES

Appendix A — Data Source Directory

Master list of named data sources referenced across all 18 in-scope sectors. For each source: the sectors that use it, the data type returned, the cost tier, and the named fallback if the primary is unavailable. Engineering teams use this list to estimate per-report data cost + identify shared infrastructure opportunities.

Source	Sectors	Returns	Cost	Fallback
Google Places + Reviews API	Universal — all sectors with physical location	Reviews + photos + hours + categories	\$0.02/lookup	Yelp Fusion API
Yelp Fusion API	Universal fallback for Google Places	Reviews + categorization + named-business intel	Free / sub	OpenStreetMap + manual
LinkedIn Sales Navigator + Apollo + ZoomInfo	All B2B sectors	B2B prospect + buyer intelligence	Subscription	LinkedIn Recruiter / direct
SimilarWeb + Semrush + Ahrefs	All sectors with online presence	Competitor traffic + SEO + paid intel	Subscription	Direct site analysis
USDA NASS	Agriculture (sector 11)	Farm-level commodity + production data	Free	State extension data
FCC Broadband Map + Form 477 + BDC	Telecom (51.C)	Broadband coverage + named competitor footprint	Free public	Direct
NTIA BEAD + USDA ReConnect + state broadband	Telecom (51.C)	Federal broadband grant programs	Free public	Direct
FDA 510(k) + PMA + DMF database	Pharma + Medical Device mfg (31-33.C)	Named regulated competitors + clearances	Free public	Direct
FDIC + NCUA + FFIEC + Call Reports	Community Banking (52.A)	Bank financial + competitor intel	Free public	Bank-direct
FINRA + SEC ADV + Form ADV Public Disclosure	Wealth Mgmt (52.C)	Adviser + firm registration + AUM	Free public	Direct
NAR + Local MLS + Zillow Group	Real Estate (53.A)	Listings + transaction data + agent intel	Free / sub	Direct

Source	Sectors	Returns	Cost	Fallback
CoStar + LoopNet + Crexi	Commercial Real Estate (53.A + 53.B)	Commercial listings + transactions + tenant intel	Subscription	Direct
FMCSA SAFER + DOT registration	Trucking (48-49.A)	Carrier license + safety + violations	Free public	Direct
DAT + Truckstop.com	Trucking (48-49.A)	Load + lane pricing + capacity intel	Subscription	Direct
SAM.gov + GSA Advantage + DLA	All federal-contractor sectors	Federal contract opportunities + spend intel	Free	Direct
ThomasNet + GlobalSpec	Manufacturing (31-33.B) + Wholesale (42)	B2B industrial sourcing + RFQ flow	Subscription	Industry Net
Xometry + Protolabs + Fictiv	Manufacturing (31-33.B)	On-demand mfg marketplaces + lead flow	Free / sub	Direct
G2 + Capterra + GetApp + Software Advice	SaaS (51.A)	B2B software reviews + category leaders	Subscription	Direct
Crunchbase + PitchBook + CB Insights	SaaS (51.A) + High-growth sectors	Funding + competitor + acquisition intel	Subscription	Public summary
AICPA + state CPA society	Accounting (54.B)	Member directory + designations	Member-only	Direct
NCARB + AIA + NSPE	A&E; (54.C)	Architect + engineer license + member intel	Member-only	State board
State Bar + Avvo + Martindale-Hubbell	Legal (54.A)	Attorney directories + reviews + practice areas	Free / sub	State bar direct
NAIS + named accreditors (NEASC, MSA, SACS)	Education (61.B)	Independent school accreditation + benchmarks	Member-only	Public summary
CIRR + Course Report + SwitchUp	Bootcamps (61.C)	Bootcamp outcomes + rankings	Free / sub	Direct
Title IV school code database + ACCSC + ACCET + COE	Trade schools (61.D)	Title IV eligibility + accreditation	Free public	Direct
Faire + Mable + Tundra	Food + Bev DTC (31-33.A) + Retail	Wholesale marketplace performance	Subscription	Direct

Source	Sectors	Returns	Cost	Fallback
UNFI + KeHE + DPI	Food + Bev DTC (31-33.A) + Wholesale (42.B)	Named distributor relationships	Direct	Trade show
NPMA + ASIS + state pest licensing boards	Pest + Security (56.C)	Industry benchmarks + license intel	Member-only	State board
NWRA + SWANA + state DEQ	Waste (56.D)	Industry benchmarks + permits	Member-only	Public
IRS PTIN + EA directory	Tax + Accounting (54.B)	Tax preparer + EA listings	Free public	Direct
Specialty Food Association + Good Food Foundation	Food + Bev DTC (31-33.A)	Industry benchmarks + trade events	Member-only	Public

Appendix B — Code Structure

The sector profile registry is a flat dictionary keyed by `profile_id`. Each profile object has the fields documented below. Engineering implements one profile at a time, passing it through the standard sector renderer to validate output before adding to production registry.

Sector profile registry structure (`sectorProfiles.js`):

```
export const SECTOR_PROFILES = {
  "hospitality.cafe_quick_service": {
    profileId: "hospitality.cafe_quick_service",
    displayName: "Cafe / Quick Service Restaurant",
    businessModel: "B2C_walk_in",
    naicsExamples: ["722513", "722515", "722511"],

    // Data layer
    dataSources: [...],          // 6+ named sources with cost + fallback
    competitorRadius: {...},     // urban / suburban / rural mi values

    // Display + analysis layer
    reviewCategories: [...],     // 8 named review categories
    opportunityPlaybook: [...], // 12-20 categories with research basis + named example
    specificityRules: [...],     // anti-hallucination rules

    // Persona + cycle layer
    personaSchema: "...",        // schema name
    personaFields: [...],        // fields to populate
    personaExample: {...},       // illustrative persona

    cycleEntries: [...],         // calendar/window/trigger/action rows

    // Cost calibration + KPIs
    costScaleBands: [...],       // 5 bands sized to sector
    kpis: [...],                 // 6-10 sector-appropriate KPIs

    // Issue + report
    issueLibrary: [...],         // collapsed-drawer issues
    reportSections: [...],       // section name + contents
  },
  "hospitality.full_service_restaurant": { ... },
  "retail.specialty_brick_mortar": { ... },
  // ... 60+ more profile_ids across 18 sectors
};
```

Appendix C — Classifier Keyword Index

Keyword index used by businessTypeKeywordRouter.js to classify free-text business type queries into sector_profile_id. Each profile_id has 8-25 trigger keywords. Keywords are grouped by specificity tier: tier-1 (unambiguous, high-confidence) and tier-2 (need additional context).

Input phrase	Routes to profile_id	Tier	Notes
coffee shop	hospitality.cafe_quick_service	T1	unambiguous tier-1
cafe	hospitality.cafe_quick_service	T1	unambiguous tier-1
barbershop	other_services.personal_care	T1	unambiguous tier-1
hair salon	other_services.personal_care	T1	unambiguous tier-1
nail salon	other_services.personal_care	T1	unambiguous tier-1
day spa	other_services.personal_care	T1	unambiguous tier-1
auto repair	other_services.auto_repair	T1	unambiguous tier-1
hvac contractor	construction.residential_trades	T1	unambiguous tier-1
electrician	construction.residential_trades	T1	unambiguous tier-1
plumber	construction.residential_trades	T1	unambiguous tier-1
roofing contractor	construction.residential_trades	T1	unambiguous tier-1
landscaper	construction.residential_trades	T1	unambiguous tier-1
general contractor	construction.commercial_gc	T2	tier-2 — could be residential too; check naics + scope
dentist	healthcare.dental_practice	T1	unambiguous tier-1
chiropractor	healthcare.allied_health	T1	unambiguous tier-1
therapist	healthcare.behavioral_health	T2	tier-2 — could be physical/mental/occupational
mental health counselor	healthcare.behavioral_health	T1	unambiguous tier-1
vet	professional.veterinary	T1	unambiguous tier-1
veterinarian	professional.veterinary	T1	unambiguous tier-1
animal hospital	professional.veterinary	T1	unambiguous tier-1
law firm	professional.legal	T1	unambiguous tier-1
cpa firm	professional.accounting	T1	unambiguous tier-1
accountant	professional.accounting	T1	unambiguous tier-1
architecture firm	professional.architecture_engineering	T1	unambiguous tier-1

Input phrase	Routes to profile_id	Tier	Notes
engineering firm	professional.architecture_engineerin g	T2	tier-2 — could be specialty engineering vs other
marketing agency	professional.consulting	T1	unambiguous tier-1
it services	professional.consulting	T2	tier-2 — could be MSP / IT consulting / other
msp	professional.consulting	T1	unambiguous tier-1 — managed services provider
real estate agent	real_estate.brokerage	T1	unambiguous tier-1
property management	real_estate.property_management	T1	unambiguous tier-1
self storage	real_estate.self_storage	T1	unambiguous tier-1
insurance agency	finance.insurance_agency	T1	unambiguous tier-1
community bank	finance.community_bank	T1	unambiguous tier-1
credit union	finance.community_bank	T1	unambiguous tier-1
financial advisor	finance.ria_wealth_management	T1	unambiguous tier-1
wealth management	finance.ria_wealth_management	T1	unambiguous tier-1
hotel	hospitality.lodging	T1	unambiguous tier-1
motel	hospitality.lodging	T1	unambiguous tier-1
bnb	hospitality.lodging	T1	unambiguous tier-1
restaurant	hospitality.full_service_restaurant	T2	tier-2 — could be cafe / QSR / FSR
bar	hospitality.bar_nightlife	T1	unambiguous tier-1
pub	hospitality.bar_nightlife	T1	unambiguous tier-1
brewery	manufacturing.food_beverage_dtc	T2	tier-2 — production vs taproom-led
coffee roaster	manufacturing.food_beverage_dtc	T1	unambiguous tier-1
cheese maker	manufacturing.food_beverage_dtc	T1	unambiguous tier-1
machine shop	manufacturing.industrial_b2b	T1	unambiguous tier-1
cnc shop	manufacturing.industrial_b2b	T1	unambiguous tier-1
contract manufacturer	manufacturing.industrial_b2b	T1	unambiguous tier-1
medical device	manufacturing.advanced_regulated	T1	unambiguous tier-1
aerospace parts	manufacturing.advanced_regulated	T1	unambiguous tier-1

Input phrase	Routes to profile_id	Tier	Notes
wholesale distributor	wholesale.durable_goods	T2	tier-2 — could be food or durable
food distributor	wholesale.nondurable_food_grocery	T1	unambiguous tier-1
trucking company	transportation.trucking_freight	T1	unambiguous tier-1
courier service	transportation.courier_last_mile	T1	unambiguous tier-1
warehouse	transportation.warehousing_3pl	T1	unambiguous tier-1
3pl	transportation.warehousing_3pl	T1	unambiguous tier-1
staffing agency	admin.staffing_agency	T1	unambiguous tier-1
janitorial	admin.facility_services	T1	unambiguous tier-1
commercial cleaning	admin.facility_services	T1	unambiguous tier-1
pest control	admin.security_pest	T1	unambiguous tier-1
security guard	admin.security_pest	T1	unambiguous tier-1
waste management	admin.waste_management	T1	unambiguous tier-1
saas company	information.saas_software	T1	unambiguous tier-1
software company	information.saas_software	T2	tier-2 — could be packaged or services
newsletter	information.media_publishing	T1	unambiguous tier-1
podcast	information.media_publishing	T1	unambiguous tier-1
isp	information.telecom_isp	T1	unambiguous tier-1
internet service provider	information.telecom_isp	T1	unambiguous tier-1
radio station	information.broadcasting_production	T1	unambiguous tier-1
tv station	information.broadcasting_production	T1	unambiguous tier-1
tutor	education.tutoring_test_prep	T1	unambiguous tier-1
test prep	education.tutoring_test_prep	T1	unambiguous tier-1
private school	education.k12_private	T1	unambiguous tier-1
montessori	education.k12_private	T1	unambiguous tier-1
coding bootcamp	education.bootcamps_online	T1	unambiguous tier-1
trade school	education.trade_technical	T1	unambiguous tier-1
cdl school	education.trade_technical	T1	unambiguous tier-1
gym	recreation.fitness_studio	T1	unambiguous tier-1

Input phrase	Routes to profile_id	Tier	Notes
yoga studio	recreation.fitness_studio	T1	unambiguous tier-1
crossfit	recreation.fitness_studio	T1	unambiguous tier-1
clothing store	retail.specialty_brick_mortar	T1	unambiguous tier-1
boutique	retail.specialty_brick_mortar	T1	unambiguous tier-1
online store	retail.ecommerce_dtc	T1	unambiguous tier-1
shopify store	retail.ecommerce_dtc	T1	unambiguous tier-1
holding company	—	OOS	OUT OF SCOPE — sector 55
government agency	—	OOS	OUT OF SCOPE — sector 92

Appendix D — Glossary

Selected terms used throughout this specification.

B2B_relationship

Business model where the customer is another business + sales cycle is long-cycle + revenue is account-based. Examples: contract manufacturer, MSP, accounting firm.

B2B_subscription

Business model where the customer is another business + revenue is recurring subscription. Examples: SaaS, MSP managed services, B2B newsletter.

B2B_transactional

Business model where the customer is another business + revenue is transaction-by-transaction (vs subscription or relationship). Examples: spot freight, wholesale distribution.

B2C_appointment

Business model where the customer is an individual + revenue is appointment-driven. Examples: dental practice, salon, vet clinic.

B2C_walk_in

Business model where the customer is an individual + revenue is walk-in. Examples: cafe, retail store, gas station.

B2C_subscription

Business model where the customer is an individual + revenue is recurring. Examples: gym membership, streaming service, premium newsletter.

Cycle entry

A sector's window/trigger/action mapping that captures when meaningful events happen (e.g. tax season for accountants, holiday season for retail, harvest season for agriculture).

Issue library

Collapsed-drawer list of common problems specific to a sector's sub-cluster, with framing for how to address each.

Master Router

Layer 0.5 of the BizRadar pipeline. Classifies inputs by NAICS-2 + business model + sector_profile_id and routes to the appropriate downstream profile.

Named entity

A specific real-world entity (cert body, software platform, federal program, association, OEM customer) referenced in an opportunity playbook entry. Named entities are verifiable at deployment time.

NAICS-2 / NAICS-4 / NAICS-6

North American Industry Classification System hierarchy. NAICS-2 = sector level (18 codes), NAICS-4 = industry group (~300 codes), NAICS-6 = industry (~1,464 codes). BizRadar v4 routes primarily by NAICS-6 with NAICS-4 + NAICS-2 fallbacks.

Opportunity playbook

A sector's 12-20 categorized growth + improvement opportunities, each with a research basis + a named real-world example.

Persona schema

The customer-type model appropriate for a sector. B2C_walk_in uses simple consumer schema; B2B_relationship uses decision-committee schema; healthcare uses patient schema.

Profile ID

A dot-notated identifier for a sub-cluster (e.g. "hospitality.cafe_quick_service"). Maps to a sector profile object in sectorProfiles.js.

Quality gate

A required condition that must pass before a phase advances. Examples: profile must have 12+ playbook entries, profile must reuse N% of substrate from existing profile, customer NPS must hold.

Specificity rule

A rule that constrains output to verified named entities + prevents hallucination. Examples: every named cert must be real, every named distributor must currently operate.

Sub-cluster

A more specific grouping within a NAICS-2 sector that has its own profile. Sector 23 (Construction) has 3 sub-clusters: residential trades, commercial GC, civil/heavy. Sector 54 has 5 sub-clusters.

End of BizRadar v4 specification

Across the 5 batches the spec covers 18 in-scope NAICS sectors with 60+ sector profiles, documents 2 out-of-scope sectors, ships a 302-code failure-rate database, defines a 5-phase implementation roadmap with explicit gates, and provides 4 appendices for engineering reference. Total addressable market: 1,464 NAICS-6 business types covered. Customer NPS improvement and revenue-attributable-to-non-hospitality-sector targets are documented in Part 4 phase gates.